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CONTEST 1 ROUND 1A

Evaluate the given limit

1 $\lim_{h \rightarrow 0} \frac{(h-5)^2 - 25}{h}$ $[(h^2 - 10h)/h = h - 10, \text{ hence limit as } h \text{ tends to } 0 \text{ is } -10]$

ANSWER: -10

2 $\lim_{h \rightarrow 0} \frac{(2+h)^2 - 4}{h}$ $[(4h + h^2)/h = 4 + h, \text{ hence limit as } h \text{ tends to } 0 \text{ is } 4]$

ANSWER: 4

3 $\lim_{h \rightarrow 0} \frac{(3-h)^2 - 9}{h}$ $[(h^2 - 6h)/h = -6 + h, \text{ hence limit as } h \text{ tends to } 0 \text{ is } -6]$

ANSWER: - 6

CONTEST 1 ROUND 1B

Find the area of a parallelogram if

1 the area is represented by x^2 , the base by $x + 3$ and the altitude by $x - 2$.

ANSWER: 36 sq. units

$[x^2 = (x + 3)(x - 2) = x^2 + x - 6, x - 6 = 0, x = 6 \text{ hence area } x^2 = 6^2 = 36]$

2 the area is represented by $x^2 - 10$, the base by x and the altitude by $x - 2$.

ANSWER: 15 sq. units

$[x^2 - 10 = x(x - 2) = x^2 - 2x, 2x = 10, x = 5, \text{ area } x^2 - 10 = 25 - 10 = 15],$

3 the area is represented by $2x^2 - 34$, the base by $x + 3$ and the altitude by $x - 3$

ANSWER: 16 sq. units

$[2x^2 - 34 = (x + 3)(x - 3) = x^2 - 9, x^2 = 34 - 9 = 25, x = 5,$

$\text{area} = 2x^2 - 34 = 2(25) - 34 = 50 - 34 = 16]$

CONTEST 1 ROUND 2: SPEED RACE

1 Solve the equation $8x^{-2/3} = 2/9$.

ANSWER: $x = 216$

[$x^{-2/3} = 1/36$, $x^{2/3} = 36$, $x^{1/3} = 6$, $x = 6^3 = 216$]

2 Two events A and B are disjoint with $P(A) = 1/3$ and $P(A \cup B) = 8/15$. Find $P(B)$.

ANSWER: $1/5$

[$P(A \cup B) = P(A) + P(B)$, $P(B) = 8/15 - 1/3 = (8 - 5)/15 = 3/15 = 1/5$]

3 Find a unit vector perpendicular to the given vector $\mathbf{a} = -6\mathbf{i} + 8\mathbf{j}$

ANSWER: $(4\mathbf{i} + 3\mathbf{j})/5$, (accept also $-(4\mathbf{i} + 3\mathbf{j})/5$)

[$\mathbf{b} = 8\mathbf{i} + 6\mathbf{j}$ is perpendicular to $\mathbf{a}$, unit vector is $(8\mathbf{i} + 6\mathbf{j})/10 = (4\mathbf{i} + 3\mathbf{j})/5$]

CONTEST 1 ROUND 5: RIDDLE

1 I am a power.

2 I am an index

3 I am an exponent

4 I am the power a number should be raised to obtain another number

5 If $a^x = y$, then $x = \log_a y$

6 I have rules unto myself such as the product rule.
Who am I?

ANSWER: LOGARITHM

CONTEST 1 ROUND 4A: TRUE OR FALSE

1 If $\log_a b = 1/c$, then $b = a^c$

ANSWER: FALSE

2 $3\log x + 1 = \log 10x^3$

ANSWER: TRUE

3 If $\log_b a = c$, then $a = b^c$

ANSWER: TRUE

CONTEST 1 ROUND 4B: TRUE OR FALSE

1 The lines with equations $x + 2y = 10$ and $2x + y = 5$ are perpendicular

ANSWER: FALSE

2 The lines with equations $x + y = 2$ and $y = 5 - x$ are parallel.

ANSWER: TRUE

3 The lines with equations $3x + y = 5$ and $x - 3y = 5$ are perpendicular.

ANSWER: TRUE

CONTEST 2 ROUND 1A

From a point P outside a circle, a tangent is drawn to touch the circle at T. From P a line is drawn to cut the circle at A and B.

1 Find the length of PT given PA = 9cm and AB = 11cm.

ANSWER: PT = $6\sqrt{5}$ cm

$$[PA = 9, PB = 9 + 11 = 20, PT^2 = PA \times PB = 9 \times 20 = 180, PT = \sqrt{180} = 6\sqrt{5}]$$

2 Find the length of PA given PT = 30cm and PB = 20cm.

ANSWER: PA = 45 cm

$$[PT^2 = PA \times PB, PA = PT^2/PB = 30^2/20 = 900/20 = 45]$$

3 Find the length of PB given PA = 18 cm, and PT = 24 cm.

ANSWER: PB = 32 cm

$$[PT^2 = PA \times PB, PB = PT^2/PA = 24 \times 24/18 = 8 \times 8/2 = 32]$$

CONTEST 2 ROUND 1B

Determine the constants A and B if

1 $(2x - 3)/(x - 2) = A + B/(x - 2)$ **ANSWER: A = 2, B = 1**

$$[2x - 3 = A(x - 2) + B, \text{ compare coefficients } A = 2, -3 = -4 + B, B = 1]$$

2 $(3x + 5)/(x + 3) = A + B/(x + 3)$ **ANSWER: A = 3, B = -4**

$$[3x + 5 = A(x + 3) + B, \text{ for } x \text{ coefficients } 3 = A, \text{ for } x^0 \quad 5 = 3A + B, \\ B = 5 - 9 = -4]$$

3 $(5x + 6)/(x - 1) = A + B/(x - 1)$ **ANSWER: A = 5, B = 11**

$$[5x + 6 = A(x - 1) + B, \text{ for } x \text{ coefficients } A = 5, \text{ for } x^0 \quad 6 = B - A, B = 5 + 6 = 11]$$

CONTEST 2 ROUND 2: SPEED RACE

1 Find the values of x for which the given function $f(x) = (x - 1)/(x^2 - 4)(x + 1)$ is not defined

ANSWER: $x = \pm 2, -1$

[denominator = $(x - 2)(x + 2)(x + 1)$, hence not defined for $x = 2, -2, -1$]

2 Find the area of the finite region bounded by the curves

$y = x^2 + 1$ and the line $y = 10$

ANSWER: 36 sq. units

[$x^2 + 1 = 10, x^2 = 9, x = \pm 3, A = \int_{-3}^3 (9 - x^2) dx = (9x - x^3/3) \Big|_{-3}^3 = (27 - 9) - (-27 + 9) = 36$]

3 State a necessary condition for the compound statement 'P and Q' to be true

ANSWER: Both P and Q must be true.

CONTEST 2 ROUND 5: RIDDLE

1 I am a rule.

2 I am used to evaluate coefficients.

3 I am normally applied to rational functions.

4 I come into play when a rational function is expressed in terms of simpler rational functions.

5 I am the object of partial fractions.

6 I involve hiding some term or covering them up.
Who am I?

ANSWER: COVER-UP RULE

CONTEST 2 ROUND 4A: TRUE OR FALSE

Given that $f(x) = (x - 4)(x^2 - 8x + 15)$

1 $f(x) = 0$ when $x = -4$

ANSWER: FALSE

2 $f(x) = 60$ when $x = 8$

ANSWER: TRUE

3 $f(x) = 1$ when $x = 5$

ANSWER: FALSE

CONTEST 2 ROUND 4B: TRUE OR FALSE

1. $-(9 - x^2) = (3 + x)(x - 3)$

ANSWER: TRUE

2. $-2x/-(6 - x) = 2x/(x - 6)$

ANSWER: FALSE

3. $(2x + 5)/(16 - x^2) = (-2x - 5)/(x^2 - 16)$

ANSWER: TRUE

CONTEST 3 ROUND 1A

Find the coordinates of the vertices of a triangle given that the sides are the lines with equations

1 $y = 0$, $x + y = 6$, $y = 2x + 6$ **ANSWER: (6, 0), (-3, 0) and (0, 6)**

[$y = 0$ and $x + y = 6$ gives (6, 0), $y = 0$ and $y = 2x + 6$ gives (-3, 0),
 $x + y = 6$ and $y = 2x + 6$ gives $3x + 6 = 6$, $x = 0$ and $y = 6$ hence (0, 6)]

2 $x = 0$, $y = 2x - 4$, $2y - x = 4$ **ANSWER: (0, -4), (0, 2) and (4, 4)**

[$x = 0$, $y = 2x - 4$ gives (0, -4), $x = 0$ and $2y - x = 4$ gives (0, 2),
 $y = 2x - 4$ and $2y - x = 4$, $2(2x - 4) - x = 4$, $3x = 12$, $x = 4$ and $y = 4$, (4, 4)]

3 $y = 0$, $y = x - 4$, $y = 2x + 4$ **ANSWER: (4, 0), (4, -2) and (0, -4)**

[$y = 0$ and $y = x - 4$ gives (4, 0), $y = 0$ and $y = 2x + 4$ gives (4, -2)
 $y = x - 4$ and $y = 2x + 4$ gives $x - 4 = 2x + 4$, $x = 0$, $y = -4$ hence (0, -4)]

CONTEST 3 ROUND 1B

A binary operation is defined on the set of integers by
 $m * n = (m - n)^2$. Find x if

1 $x * 4 = 1$

ANSWER: $x = 5, 3$

[$m * n = (m - n)^2$, $(x * 4) = 1$, $(x - 4)^2 = 1$, $x = 4 \pm 1$, $x = 5, 3$]

2 $x * (-3) = 9$

ANSWER: $x = 0, -6$

[$x * (-3) = 9$, $(x + 3)^2 = 9$, $x + 3 = \pm 3$, $x = -3 \pm 3$, $x = 0, -6$]

3 $x * 5 = 16$

ANSWER: $x = 9, 1$

[$(x - 5)^2 = 16$, $x - 5 = \pm 4$, $x = 5 \pm 4$, $x = 9, 1$]

CONTEST 3 ROUND 2: SPEED RACE

1 Find the coordinates of the stationary points of $y = x^3 - 9x^2 + 24x$

ANSWER: (2, 20), (4, 16)

$$[dy/dx = 3x^2 - 18x + 24 = 3(x^2 - 6x + 8) = 3(x - 2)(x - 4) = 0, x = 2, 4]$$

$$\text{For } x = 2, y = 8 - 36 + 48 = 20, \text{ for } x = 4, y = 64 - 144 + 96 = 16]$$

2 Find an equivalent but simpler trigonometric expression for $\cos(x + \pi/2)$

ANSWER: $-\sin x$

$$[\cos(x + \pi/2) = \cos x \cos(\pi/2) - \sin x \sin(\pi/2) = -\sin x]$$

3 If $6x - 9 \leq f(x) \leq x^2$ for all x , find $\lim_{x \rightarrow 3} f(x)$

ANSWER: 9

$$[\text{Take limit as } x \text{ tends to } 3, 9 \leq \lim_{x \rightarrow 3} f(x) \leq 9, \text{ hence limit} = 9]$$

CONTEST 3 ROUND 5: RIDDLE

1 I am a geometrical figure.

2 I am a polygon.

3 I am a symmetrical figure but I am neither equilateral nor equiangular.

4 My diagonals produce two pairs of congruent isosceles triangles.

5 I have only one axis of symmetry.

6 I fly and I can be flown.

Who am I?

ANSWER: KITE

CONTEST 3 ROUND 4A: TRUE OR FALSE

The function $f(x) = x^3 - 3x^2 + 2x - 6$ has the following as a factor

1 $(x - 3)$

ANSWER: TRUE

2 $(x - 1)$

ANSWER: FALSE

3 $(x - 2)$

ANSWER: FALSE

CONTEST 3 ROUND 4B: TRUE OR FALSE

1 $\sin 150^\circ = \frac{1}{2}$

ANSWER: TRUE

2 $\cos 150^\circ = \frac{\sqrt{3}}{2}$

ANSWER: FALSE

3 $\tan 150^\circ = \frac{1}{\sqrt{3}}$

ANSWER: FALSE

CONTEST 4 ROUND 1A

The probabilities that Kwame, Edem and Adamu will score in a penalty shoot-out are $\frac{4}{5}$, $\frac{2}{3}$ and $\frac{3}{4}$ respectively. Find the probability that if they participate in a penalty shoot-out

1 none of the three will score,

ANSWER: $\frac{1}{60}$

$$[(1 - \frac{4}{5})(1 - \frac{2}{3})(1 - \frac{3}{4}) = (\frac{1}{5})(\frac{1}{3})(\frac{1}{4}) = \frac{1}{60}]$$

2 only Kwame does not score,

ANSWER : $\frac{1}{10}$

$$[(1 - \frac{4}{5})(\frac{2}{3})(\frac{3}{4}) = (\frac{1}{5})(\frac{2}{3})(\frac{3}{4}) = \frac{6}{60} = \frac{1}{10}]$$

3 all three score.

ANSWER: $\frac{2}{5}$

$$[(\frac{4}{5})(\frac{2}{3})(\frac{3}{4}) = \frac{24}{60} = \frac{2}{5}]$$

CONTEST 4 ROUND 1B

Express y in terms of x if

$$1 \quad (\frac{3}{2})^y = (\frac{8}{27})^x$$

ANSWER: $y = -3x$

$$[(\frac{3}{2})^y = (\frac{27}{8})^{-x} = (\frac{3}{2})^{-3x}, \text{ hence } y = -3x]$$

$$2 \quad (25)^{(y-3)} = 5^{-3(2x-4)}$$

ANSWER: $y = -3x + 9$

$$[5^{2(y-3)} = 5^{-3(2x-4)}, 2y - 6 = -6x + 12, y = -3x + 9]$$

$$3 \quad (81)^{2x+1} = (27)^{4y-2}$$

ANSWER: $y = \frac{2x}{3} + \frac{5}{6}$

$$[3^{4(2x+1)} = 3^{3(4y-2)}, 12y - 6 = 8x + 4, 12y = 8x + 10, y = \frac{2x}{3} + \frac{5}{6}]$$

CONTEST 4 ROUND 4A: TRUE OR FALSE

For the function $y = 2(x + 3)^2 - 5$,

1 the vertex is (3, -5)

ANSWER: FALSE

2 the number 3 indicates a horizontal translation to the right,

ANSWER: FALSE

3 the minimum value is -5

ANSWER: TRUE

CONTEST 4 ROUND 4B: TRUE OR FALSE

1 The graphs of the functions $y = \sin x$ and $y = \cos x$ have the same shape.

ANSWER: TRUE

2 The graph of $y = \sin x$ is a horizontal translation to the right by $\pi/2$ of the graph of $y = \cos x$.

ANSWER: TRUE

3 The graph of $y = \sin x$ repeats itself after every interval of length π radians

ANSWER: FALSE

CONTEST 4 ROUND 5: RIDDLE

1 I am an equation.

2 To be precise, I am a quadratic equation.

3 I have in reality one root.

4 This means my quadratic expression is an exact square.

5 The sum of my roots is 12.

6 The product of my roots is an exact square.
Who am I?

ANSWER: $(x - 6)^2 = 0$, or $x^2 - 12x + 36 = 0$

CONTEST 4 ROUND 2: SPEED RACE

1 Vectors **a**, **b** are given by $\mathbf{a} = 5\mathbf{i} + \mathbf{j}$ and $\mathbf{b} = \mathbf{i} + 5\mathbf{j}$, express the vector $(\mathbf{a} + \mathbf{b})$ in a bearing form.

ANSWER: $(6\sqrt{2}, 045^\circ)$

$[\mathbf{a} + \mathbf{b} = 6\mathbf{i} + 6\mathbf{j} = 6(\mathbf{i} + \mathbf{j}), |\mathbf{a} + \mathbf{b}| = 6\sqrt{2}, \text{direction is } 045^\circ, \text{ hence } (6\sqrt{2}, 045^\circ)]$

2 A linear transformation is defined by $T: (x, y) \rightarrow (y, -x)$. Find the coordinates of the image of the point $A(2, -3)$ under T .

ANSWER: $(-3, -2)$

$[(2, -3) \rightarrow (-3, -2)]$

3 Find the equation of a curve with gradient $dy/dx = x^2 - 2x$ and passing through the point $A(3, 3)$.

ANSWER: $y = x^3/3 - x^2 + 3$

$[y = x^3/3 - x^2 + C, 3 = 9 - 9 + C, C = 3, \text{ hence } y = x^3/3 - x^2 + 3]$

CONTEST 5 ROUND 1A

Find the value of a if

$$1 \quad (3 + a\sqrt{2})(3 - 3\sqrt{2}) = -3 - 3\sqrt{2} \quad \text{ANSWER: } a = 2$$
$$[9 - 6a + (-9 + 3a)\sqrt{2} = -3 - 3\sqrt{2}, 9 - 6a = -3, 12 = 6a, a = 2, -9a + 3a = -3, 3a = 6, a = 2]$$

$$2 \quad (a - \sqrt{3})(3 + 2\sqrt{3}) = 6 + 5\sqrt{3} \quad \text{ANSWER: } a = 4$$
$$[3a - 6 + (2a - 3)\sqrt{3} = 6 + 5\sqrt{3}, 3a - 6 = 6, a = 4, 2a - 3 = 5, a = 4]$$

$$3 \quad (4 + 2\sqrt{5})(3 + a\sqrt{5}) = -8 - 2\sqrt{5} \quad \text{ANSWER: } a = -2$$
$$[12 + 10a + (4a + 6)\sqrt{5} = -8 - 2\sqrt{5}, 12 + 10a = -8, a = -2, 4a + 6 = -2, a = -2]$$

CONTEST 5 ROUND 1B

Solve the trigonometric equation for values in the interval $0 \leq x \leq \pi$

$$1 \quad \sin^2 x - 3\sin x + 2 = 0 \quad \text{ANSWER: } x = \pi/2 \text{ radians}$$
$$[\sin^2 x - 3\sin x + 2 = (\sin x - 2)(\sin x - 1) = 0, \sin x = 1, x = \pi/2, \sin x = 2 \text{ no solution}]$$

$$2 \quad \cos^2 x + 2\cos x + 1 = 0 \quad \text{ANSWER: } x = \pi \text{ radians}$$
$$[\cos^2 x + 2\cos x + 1 = (\cos x + 1)^2, \cos x = -1, x = \pi]$$

$$3 \quad 2\tan^2 x = 1 \quad \text{ANSWER: } x = \pi/4, 3\pi/4 \text{ radians}$$
$$[\tan^2 x = 1/2, \tan x = \pm 1/\sqrt{2}, \tan x = 1/\sqrt{2}, x = \pi/4, \tan x = -1/\sqrt{2}, x = 3\pi/4]$$

CONTEST 5 ROUND 2: TRUE OR FALSE

1 Express $\log(125/128)$ in terms of $\log 2$

ANSWER: $3 - 10\log 2$

$$[125/128 = 5^3/2^7 = 5^3 2^3/2^{10} = 1000/2^{10}, \log(125/128) = 3 - 10\log 2]$$

2 Simplify the rational expression $(x^2 - x - 2)/(x^2 - 5x + 6)$

ANSWER: $(x + 1)/(x - 3)$

$$[(x - 2)(x + 1)/(x - 3)(x - 2) = (x + 1)/(x - 3)]$$

3 Find the solution set of the inequality $x^3 + 5x^2 + 2x + 10 < 0$

ANSWER: $\{x : x < -5\}$

$$[x^2(x + 5) + 2(x + 5) < 0, (x^2 + 2)(x + 5) < 0, x < -5]$$

CONTEST 5 ROUND 5: RIDDLE

1 I am a curve in the cartesian plane

2 I am the simplest such curve.

3 With me, it is no curve and no bend.

4 I am a curve with a uniform gradient

5 I offer the shortest distance between two points in the plane.

6 My equation can be put in the form $y = ax + b$

Who am I?

ANSWER: STRAIGHT LINE

CONTEST 5 ROUND 4A: TRUE OR FALSE

1 Opposite angles of a cyclic quadrilateral add up to a straight angle

ANSWER: TRUE

2 An exterior angle of a cyclic quadrilateral is congruent to the interior opposite angle

ANSWER: TRUE

3 When two circles are tangent internally, they have exactly one common tangent.

ANSWER: TRUE

CONTEST 5 ROUND 4B: TRUE OR FALSE

Consider the curve with equation $y = (x - 4)(x + 6)$

1 The point (5, 11) is on the curve.

ANSWER: TRUE

2 The point (-2, -20) is on the curve.

ANSWER: FALSE

3 The point (6, 24) is on the curve.

ANSWER: TRUE

CONTEST 6 ROUND 1A

P and Q are simple statements.

State the condition for the given compound statement to be false.

1 P implies Q

ANSWER: P is true and Q false

2 P and Q

ANSWER: One of P, Q is false

3 P or Q

ANSWER: Both P and Q are false

CONTEST 6 ROUND 1B

Find the matrix of the given linear transformation

1 T: $(x, y) \rightarrow (x - 2y, x + 3y)$

ANSWER: $\begin{pmatrix} 1 & -2 \\ 1 & 3 \end{pmatrix}$

2 T: $(x, y) \rightarrow (-x + 2y, 3x - 5y)$

ANSWER: $\begin{pmatrix} -1 & 2 \\ 3 & -5 \end{pmatrix}$

3 T: $(x, y) \rightarrow (2y, x - 2y)$

ANSWER: $\begin{pmatrix} 0 & 2 \\ 1 & -2 \end{pmatrix}$

CONTEST 6 ROUND 2: SPEED RACE

1 Find the area of the finite region bounded by the curves $f(x) = 2x$ and $g(x) = x^2$

ANSWER: $4/3$

$$[2x = x^2, x = 0, 2; \text{ Area} = \int_0^2 (2x - x^2) dx = x^2 - x^3/3 \Big|_0^2 = 4 - 8/3 = 4/3]$$

2 Factorise completely $128x^3 - 54y^3$

ANSWER: $2(4x - 3y)(16x^2 + 12xy + 9y^2)$

$$[2(64x^3 - 27y^3) = 2((4x)^3 - (3y)^3) = 2(4x - 3y)(16x^2 + 12xy + 9y^2)]$$

3 Determine the range of the function $y = -3x^2 + 12x - 10$

ANSWER: $\{y: y \leq 2\}$

$$[y = -3(x^2 - 4x) - 10 = -3(x^2 - 4x + 4) + 12 - 10 = -3(x - 2)^2 + 2, y \leq 2]$$

CONTEST 6 ROUND 5: RIDDLE

1 I am a polygon.

2 When I am equiangular, I am not equilateral.

3 In my regular form, I fit my outline in 20 positions.

4 In my regular form I have a center of symmetry.

5 In my regular form I have 10 axes of symmetry.

6 In my regular form an interior angle has a measure of 144° .

Who am I?

ANSWER: DECAGON

CONTEST 6 ROUND 4A: TRUE OR FALSE

1 A triangle with angles in a ratio of 1 : 1 : 2 is a right angle triangle.

ANSWER: TRUE

2 A triangle with angles in a ratio of 1 : 2 : 3 is a right angle triangle.

ANSWER: TRUE

3 A triangle with angles in a ratio of 1 : 3 : 5 is a right angle triangle.

ANSWER: FALSE

CONTEST 6 ROUND 4B: TRUE OR FALSE

1 $\sin 25^\circ = \sin 155^\circ$

ANSWER: TRUE

2 $\cos 36^\circ = \cos 144^\circ$

ANSWER: FALSE

3 $\tan 53^\circ = \tan 127^\circ$

ANSWER: FALSE

CONTEST 7 ROUND 1A

Find functions $g(x)$ and $h(x)$ such that $f(x) = g(h(x))$ given

1 $f(x) = (3x - 2)^2$

ANSWER: $g(x) = x^2$ and $h(x) = 3x - 2$

2 $f(x) = 4/(2x + 5)^3$

ANSWER: $g(x) = 4/x^3$, $h(x) = (2x + 5)$

3 $f(x) = \sqrt{7 - x^2}$

ANSWER: $g(x) = \sqrt{x}$, $h(x) = 7 - x^2$

CONTEST 7 ROUND 1B

Find values of b and c such that the quadratic inequality

1 $x^2 + bx + c > 0$ has solution set $\{x: x > 3, \text{ or } x < -3\}$

ANSWER: $b = 0$, $c = -9$

$[(x - 3)(x + 3) = x^2 - 9 > 0, \text{ hence } b = 0, c = -9]$

2 $x^2 + bx + c < 0$ has solution set $\{x: 3 < x < 5\}$

ANSWER: $b = -8$, $c = 15$

$[(x - 3)(x - 5) = x^2 - 8x + 15 < 0, b = -8, c = 15]$

3 $x^2 + bx + c > 0$ has solution set $\{x: x < 3, \text{ or } x > 6\}$

ANSWER: $b = -9$, $c = 18$

$[(x - 3)(x - 6) > 0, x^2 - 9x + 18 > 0, b = -9, c = 18]$

CONTEST 7 ROUND 2: SPEED RACE

1 Solve the equation $1/f = 1/u + 1/v$ for v

ANSWER: $v = uf/(u - f)$

[$1/v = 1/f - 1/u = (u - f)/uf$, $v = uf/(u - f)$]

2 Simplify $3\sqrt{8} - \sqrt{32} + 3\sqrt{72} + \sqrt{75}$

ANSWER: $20\sqrt{2} + 5\sqrt{3}$

[$6\sqrt{2} - 4\sqrt{2} + 18\sqrt{2} + 5\sqrt{3} = 20\sqrt{2} + 5\sqrt{3}$]

3 Find the domain of the function $f(x) = \sqrt[3]{x - 2}$

ANSWER: {x: x is a real number} or {real numbers}

CONTEST 7 ROUND 5: RIDDLE

1 In Mathematics, I may be a statement requiring a proof.

2 I may also be a question requiring a method of solution.

3 I am more than an equation or an inequality.

4 Life is full of me.

5 I am that which is to be solved.

6 If you still cannot solve the problem then you have a problem.
Who am I?

ANSWER: PROBLEM

CONTEST 7 ROUND 4A: TRUE OR FALSE

If $ax^2 - bx + c = 0$, then

1 $x = (-b \pm \sqrt{b^2 - 4ac})/2a$

ANSWER: FALSE

2 $x = (-b \pm \sqrt{b^2 + 4ac})/2a$

ANSWER: FALSE

3 $x = (b \pm \sqrt{b^2 - 4ac})/2a$

ANSWER: TRUE

CONTEST 7 ROUND 4B: TRUE OR FALSE

The given set of ordered pairs is a function

1 $A = \{(1, 3), (1, 2), (2, 3), (2, 5)\}$

ANSWER: FALSE [many to many relation is not a function]

2 $B = \{(2, 1), (3, 1), (4, 1), (5, 1)\}$

ANSWER: TRUE [many to one relation is a function]

3 $C = \{(1, 3), (2, 5), (3, 4), (4, 7)\}$

ANSWER: TRUE [one to one relation is a function]

CONTEST 8 ROUND 1A

Determine the solution set of the given equation

1 $|x| = x$ (NB: Read as absolute value of $x = x$)

ANSWER: $\{x: x = 0\}$

[$x = x$ for $x = 0$, $x = -x$ for $x = 0$, hence $\{x: x = 0\}$]

2 $|x| + x = 4$

ANSWER: $\{2\}$ or $\{x: x = 2\}$

[$|x| = (4 - x)$, $x^2 = 16 - 8x + x^2 = 0$, $8x = 16$, $x = 2$]

3 $|x + 1| = x + 1$

ANSWER: $\{x: x = -1\}$

[$|x + 1| = x + 1$ for $x + 1 = 0$, ie for $x = -1$, hence $\{x: x = -1\}$]

CONTEST 8 ROUND 1B

Find the sum to infinity of the exponential series

1. $8 - 4 + 2 - 1 + 1/2 + \dots$

ANSWER: $16/3$

[GP, $a = 8$, $r = -1/2$, $S_\infty = a/(1 - r) = 8/(1 - (-1/2)) = 8/(3/2) = 16/3$]

2. $6 + 3 + 3/2 + 3/4 + 3/8 + \dots$

ANSWER: 12

[GP, $a = 6$, $r = 1/2$, $S_\infty = a/(1 - r) = 6/(1 - 1/2) = 6/(1/2) = 6(2) = 12$]

3. $10 - 2 + 2/5 - 2/25 + 2/125 + \dots$

ANSWER: $25/3$

[GP, $a = 10$, $r = -1/5$, $S_\infty = a/(1 - r) = 10/(1 + 1/5) = 10/(6/5) = 50/6 = 25/3$]

CONTEST 8 ROUND 2: SPEED RACE

1 Differentiate the function $y = (2x - 2)/\sqrt{x}$. Simplify your answer.

ANSWER: $dy/dx = \sqrt{x}/x + \sqrt{x}/x^2$

$$[y = 2\sqrt{x} - 2/\sqrt{x}, dy/dx = 1/\sqrt{x} + 1/x\sqrt{x} = \sqrt{x}/x + \sqrt{x}/x^2]$$

2 Find an equation for the function $g(x)$ which has the graph of $f(x) = x^2$, but moved 3 units to the right and 4 units vertically up.

ANSWER: $g(x) = (x - 3)^2 + 4$ $[g(x) = f(x - 3) + 4 = (x - 3)^2 + 4]$

3 Rationalize the denominator $(3\sqrt{5} - 5\sqrt{3})/(3\sqrt{5} + 5\sqrt{3})$

ANSWER: $-4 + \sqrt{15}$

$$[(3\sqrt{5} - 5\sqrt{3})^2/(45 - 75) = [(45 + 75) - 30\sqrt{15}]/-30 = -4 + \sqrt{15}]$$

CONTEST 8 ROUND 5: RIDDLE

1 I am a binary operation on the set of real numbers.

2 I am one of the four basic operations on the real numbers.

3 I am neither associative nor commutative.

4 I have no identity.

5 With me, there is a minuend and a subtrahend.

6 I result in a difference.

Who am I?

ANSWER: SUBTRACTION

CONTEST 8 ROUND 4A: TRUE OR FALSE

1 The equation $x^2 = 18$ is equivalent to $x = 3\sqrt{2}$

ANSWER: FALSE

2 The equation $x^3 = 54$ is equivalent to $x = 3\sqrt[3]{2}$.

ANSWER: TRUE

3 The equation $x^4 = 16$ is equivalent to $x = \pm 2$.

ANSWER: TRUE

CONTEST 8 ROUND 4B: TRUE OR FALSE

In the following a, b, and c are real numbers.

1 If $a/c + b/c = (a + b)/c$, then $a/b + a/c = a/(b + c)$

ANSWER: FALSE

2 If $a/b + a/c = a/(b + c)$, then $a/c + b/c = (a + b)/c$

ANSWER: TRUE

3 If $a/c - b/c = (a - b)/c$, then $a/b - a/c = a/(b - c)$.

ANSWER: FALSE

CONTEST 9 ROUND 1A

Find the sum and simplify your answer

1 $4\sqrt[3]{40} + 2\sqrt[3]{625} - 3\sqrt[3]{320}$

ANSWER: $6\sqrt[3]{5}$

$[4(2)\sqrt[3]{5} + 2(5)\sqrt[3]{5} - 3(4)\sqrt[3]{5} = (8 + 10 - 12)\sqrt[3]{5} = 6\sqrt[3]{5}]$

2 $\frac{2}{3}\sqrt{18} - \frac{\sqrt{50}}{5} + \frac{3}{4}\sqrt{32}$

ANSWER: $4\sqrt{2}$

$[\frac{2}{3}(3\sqrt{2}) - \frac{5\sqrt{2}}{5} + \frac{3}{4}(4\sqrt{2}) = (2 - 1 + 3)\sqrt{2} = 4\sqrt{2}]$

3 $4\sqrt{12} - 3\sqrt{27} + \frac{\sqrt{108}}{6}$

ANSWER: 0

$[4(2)\sqrt{3} - 3(3)\sqrt{3} + (6/6)\sqrt{3} = (8 - 9 + 1)\sqrt{3} = 0]$

Commented [D1]:

CONTEST 9 ROUND 1B

Find the coordinates of the stationary points of the given function

1 $y = x - 4/x^2$

ANSWER: (-2, -3)

$[dy/dx = 1 + 8/x^3 = 0, x^3 = -8, x = -2, y = -2 - 4/4 = -3, \text{ hence } (-2, -3)]$

2 $y = x^2(x^2 - 2)$

ANSWER: (0, 0), (1, -1), (-1, -1)

$[y = x^4 - 2x^2, dy/dx = 4x^3 - 4x = 4x(x^2 - 1) = 0, x = 0, 1, -1, \text{ hence } (0, 0) (1, -1) (-1, -1)]$

3 $y = x + 9/x$

ANSWER: (3, 6) (-3, -6)

$[dy/dx = 1 - 9/x^2, x^2 = 9, x = 3, -3, \text{ for } x = 3, y = 3 + 3 = 6, (3, 6) \text{ and for } x = -3, y = -3 - 3 = -6, (-3, -6)]$

CONTEST 9 ROUND 2: SPEED RACE

1 Simplify $(x^2 - x - 6)/(x^2 - 2x - 8)$

ANSWER: $(x - 3)/(x - 4)$

$$[(x + 2)(x - 3)/(x - 4)(x + 2) = (x - 3)/(x - 4)]$$

2 Find the area of the finite region bounded by the curves $y = 9 - x^2$ and $y = 9 - 3x$ **ANSWER: $9/2$** $[9 - x^2 = 9 - 3x, x^2 = 3x, x = 0, 3,$

$$\text{Area} = \int_0^3 (3x - x^2) dx = 3x^2/2 - x^3/3 \Big|_0^3 = 27/2 - 27/3 = 9/2]$$

3 Identify the polygon whose exterior angle in its regular form is $\pi/5$ radians

ANSWER: DECAGON

$$[2\pi/n = \pi/5, n = 2(5) = 10, \text{ hence a decagon}]$$

CONTEST 9 ROUND 5: RIDDLE

1 I am a statistical measure.

2 I am never negative.

3 Even though I am a mean, I am used as a measure of dispersion.

4 I am an average of deviations.

5 However there is nothing standard about me.

6 I am a mean of the absolute deviations from the mean.

Who am I?

ANSWER: MEAN DEVIATION

CONTEST 9 ROUND 4A: TRUE OR FALSE

1 The solution set of $x > a$, or $x < a$ is the set of real numbers.

ANSWER: FALSE

2 The solution set of $x > a$ and $x < a$ is the empty set.

ANSWER: TRUE

3 The solution set of the equation $|x| = 0$ is the empty set.

ANSWER: FALSE

CONTEST 9 ROUND 4B: TRUE OR FALSE

1 The line through the points $(5, -3)$ and $(-5, -3)$ has an undefined slope.

ANSWER: FALSE

2 The line through the points $(1, 2)$ and $(1, -2)$ has an undefined slope.

ANSWER: TRUE

3 The straight line intersecting the y-axis at 3 and the x-axis at 3 has slope of 1.

ANSWER: FALSE

CONTEST 10 ROUND 1A

Express as a single logarithm and simplify as much as possible.

$$1 \log(x^3 - y^3) - \log(x - y)$$

ANSWER: $\log(x^2 + xy + y^2)$

$$[\log[(x^3 - y^3)/(x - y)] = \log(x^2 + xy + y^2)]$$

$$2 \log(x^2 + y^2) + \log(x^2 - y^2)$$

ANSWER: $\log(x^4 - y^4)$

$$[\log(x^2 + y^2)(x^2 - y^2) = \log(x^4 - y^4)]$$

$$3 \log(x^2 - xy + y^2) + \log(x + y)$$

ANSWER: $\log(x^3 + y^3)$

$$[\log(x^2 - xy + y^2)(x + y) = \log(x^3 + y^3)]$$

CONTEST 10 ROUND 1B

A binary operation is defined on the set Z of integers by $a * b = a^2 - 3ab$ for $a, b \in Z$. Solve for x from the given equation

$$1 \ x * 3 = 10$$

ANSWER: $x = 10, -1$

$$[x^2 - 3(3x) = 10, x^2 - 9x - 10 = (x - 10)(x + 1) = 0, x = 10, -1]$$

$$2 \ x * (-3) = -20$$

ANSWER: $x = -5, -4$

$$[x^2 + 3(3x) = -20, x^2 + 9x + 20 = (x + 5)(x + 4) = 0, x = -5, -4]$$

$$3 \ x * x = -32$$

ANSWER: $x = \pm 4$

$$[x^2 - 3x^2 = -32, 2x^2 = 32, x^2 = 16, x = \pm 4]$$

ROUND 2: SPEED RACE

1 Find the value(s) of x if the matrix $A = \begin{pmatrix} 2x & 2 \\ 4 & x \end{pmatrix}$ has no inverse.

ANSWER: $x = \pm 2$

[$\det A = 2x^2 - 8 = 0$, $x^2 = 4$, $x = \pm 2$]

2 Find the solution set of the absolute value inequality $|2x - 3| < 5$

ANSWER: $\{x : -1 < x < 4\}$

$[-5 < 2x - 3 < 5, -2 < 2x < 8, -1 < x < 4]$

3 Find the point of inflexion of the curve $y = x^3 - 6x^2 + 9x$

ANSWER: (2, 2) [$dy/dx = 3x^2 - 12x + 9$, $d^2y/dx^2 = 6x - 12 = 0$, $x = 2$,
 $y = 8 - 24 + 18 = 26 - 24 = 2$, hence (2, 2)]

CONTEST 10 ROUND 5: RIDDLE

1 I am a 4 – digit number

2 My digits are distinct even integers.

3 The number formed by my first two digits is a square.

4 My last digit is a cube.

5 My third digit is a prime.

6 The number formed by my first two digits is the square of my last digit.

Who am I?

ANSWER: 6428

CONTEST 10 ROUND 4A: TRUE OR FALSE

1 The line with equation $3y = 5x + 2$ has a y-intercept of 2

ANSWER: FALSE

2 The line with equation $2y = 3x - 4$ has an x- intercept of - 4

ANSWER: FALSE

3 The line with equation $y = 2x - 1$ has a y-intercept of -1

ANSWER: TRUE

CONTEST 10 ROUND 4B: TRUE OR FALSE

1 $\sqrt[5]{-32}$ is not a real number.

ANSWER: FALSE

.

2 $\sqrt[4]{-64}$ is a real number.

ANSWER: FALSE

3 $\sqrt[3]{-12}$ is a real number.

ANSWER: TRUE

CONTEST 11 ROUND 1A

Simplify the expression $(x + 1/x)$ if

1 $x = 2 - \sqrt{3}$

ANSWER: 4

$$[2 - \sqrt{3} + 1/(2 - \sqrt{3}) = (2 - \sqrt{3}) + (2 + \sqrt{3}) = 4]$$

2 $x = 3 + \sqrt{8}$

ANSWER: 6

$$[3 + \sqrt{8} + 1/(3 + \sqrt{8}) = (3 + \sqrt{8}) + (3 - \sqrt{8}) = 6]$$

3 $x = 5 - \sqrt{26}$

ANSWER: $-2\sqrt{26}$

$$[5 - \sqrt{26} + 1/(5 - \sqrt{26}) = (5 - \sqrt{26}) + (5 + \sqrt{26})/-1 = -2\sqrt{26}]$$

CONTEST 11 ROUND 1B

Find the next three terms of the sequence defined by the recurrence relation

1 $U_{n+1} = 2U_n + 2n, U_1 = 3$

ANSWER: $U_2 = 8, U_3 = 20, U_4 = 46$ $[U_2 = 2U_1 + 2(1) = 2(3) + 2 = 8,$

$$U_3 = 2U_2 + 2(2) = 2(8) + 4 = 20, U_4 = 2U_3 + 2(3) = 2(20) + 6 = 46]$$

2 $U_{n+1} = 5 - U_n, U_1 = 2$

ANSWER: $U_2 = 3, U_3 = 2, U_4 = 3$

$$[U_2 = 5 - U_1 = 5 - 2 = 3, U_3 = 5 - U_2 = 5 - 3 = 2, U_4 = 5 - U_3 = 5 - 2 = 3]$$

3 $U_{n+1} = 7 - 2U_n, U_1 = -3$

ANSWER: $U_2 = 13, U_3 = -19, U_4 = 45$ $[U_2 = 7 - 2U_1 = 7 - 2(-3) = 13,$

$$U_3 = 7 - 2(13) = -19, U_4 = 7 - 2U_3 = 7 - 2(-19) = 45]$$

CONTEST 11 ROUND 2: SPEED RACE

1 Find the area of a rhombus if the diagonals have lengths 25 cm and 20 cm

ANSWER: 250 cm²

[Area = $\frac{1}{2} d_1 \times d_2 = \frac{1}{2} (25 \times 20) = 250 \text{ cm}^2$]

2 Solve for y in terms of x from the proportion $2x : y = y : 18x$

ANSWER: $y = \pm 6x$

[$2x/y = y/18x$, $36x^2 = y^2$, $y = \pm 6x$]

3 Find the value of x if the vector $x\mathbf{i} + 3\mathbf{j}$ is parallel to the vector $2\mathbf{i} + 12\mathbf{j}$.

ANSWER: $x = \frac{1}{2}$

[$x/2 = 3/12$, $x = 6/12 = \frac{1}{2}$]

CONTEST 11 ROUND 5: RIDDLE

1 I am a statistical measure.

2 I am concerned with variability in a given data set.

3 I cover a range of statistical quantities.

4 Examples of me are the range and mean deviation.

5 Others are variance and standard deviation.

6 It appears I am involved somehow with dispersion.

Who am I?

ANSWER: MEASURE OF DISPERSION

CONTEST 11 ROUND 4: TRUE OR FALSE

1 $x^3 - 3x^2 + 5x - 15 = (x - 3)(x^2 + 5)$

ANSWER: TRUE

2 $x^3 + 2x^2 - 4x - 8 = (x + 2)(x - 2)^2$

ANSWER: FALSE

3 $x^3 + 2x^2 - 2x - 8 = (x - 2)(x^2 + 4)$

ANSWER: FALSE

CONTEST 11 ROUND 4B: TRUE OR FALSE

1 The domain of the function $f(x) = \sqrt[3]{(x - 2)}$ is the set $\{x: x \geq 2\}$

ANSWER: FALSE

2 The domain of the function $f(x) = \sqrt{(2 - x)}$ is the set $\{x: x \leq 2\}$

ANSWER: TRUE

3 The domain of the function $f(x) = 1/\sqrt{(x - 3)}$ is the set $\{x: x \geq 3\}$

ANSWER: FALSE

CONTEST 12 ROUND 1A

Write an inequality which defines the given region in the cartesian x - y plane

1 above the line with equation $x + y = 5$,

ANSWER: $y > 5 - x$ or $x + y > 5$

[$y = 5 - x$, above the line is $y > 5 - x$ or $x + y > 5$]

2 below the line with equation $2x - y = 3$,

ANSWER: $y < 2x - 3$ or $2x - y > 3$

[$y = 2x - 3$, below the line is $y < 2x - 3$ or $2x - y > 3$]

3 above the line $3x - y = 7$

ANSWER: $y > 3x - 7$, or $3x - y < 7$

[$y = 3x - 7$, above the line is $y > 3x - 7$, or $3x - y < 7$]

CONTEST 12 ROUND 1B

1 Find the length h of the altitude to the base of an isosceles triangle given the base has length 10 cm and the congruent sides have length 13 cm

ANSWER: 12 cm

[$h^2 + 5^2 = 13^2$, $h^2 = 169 - 25 = 144$, $h = 12$ cm]

2 Find the length of the base of an isosceles triangle given the congruent sides have length 15 cm and the altitude to the base has length 12 cm

ANSWER: 18 cm

[$x^2 + 12^2 = 15^2$, $x^2 = 225 - 144 = 81$, $x = 9$, base = $2x = 2(9) = 18$]

3 Find the length of the congruent sides of an isosceles triangle with base of length 30 cm and altitude of length 8 cm to the base.

ANSWER: 17 cm

[$x^2 = 15^2 + 8^2 = 225 + 64 = 289$, $x = 17$]

CONTEST 12 ROUND 2: SPEED RACE

1 A circle has radius 120 cm. An arc has length 132 cm. Find the angle subtended by the arc at the center in radians.

ANSWER: 1.1 radians

[$s = r\theta$, $\theta = s/r = 132/120 = 1.1$ radians]

2 Simplify $\cos^2 x / (\cos x - \sin x)$ (Assume $\cos x \neq \sin x$)

ANSWER: $\cos x + \sin x$

[$(\cos^2 x - \sin^2 x) / (\cos x - \sin x) = (\cos x + \sin x)(\cos x - \sin x) / (\cos x - \sin x) = \cos x + \sin x$]

3 Find an equivalent cartesian equation in x and y for a curve defined by

$x = 4t^2$, $y = 2t$

ANSWER: $y^2 = x$ [$x = 4t^2 = (2t)^2 = y^2$]

CONTEST 12 ROUND 5: RIDDLE

1 There are three of us.

2 One of us is the median.

3 However the other two are not the mean or the mode.

4 We are ranked according to some order as first, second, etc

5 We can be obtained from the cumulative frequency diagram.

6 We divide a distribution into four equal parts.

Who are we?

ANSWER: QUARTILES

CONTEST 12 ROUND 4A: TRUE OR FALSE

1 The interior angle of a pentagon has measure 72° .

ANSWER: FALSE [only if it is regular]

2 The exterior angle of a regular octagon has measure 45° .

ANSWER: TRUE

3 The interior angle of a regular decagon has measure 140° .

ANSWER: FALSE

CONTEST 12 ROUND 4B: TRUE OR FALSE

1 The graph of any line in the $x - y$ plane defines y as a function of x .

ANSWER: FALSE

2 A function has an inverse if it is many to one.

ANSWER: FALSE

3 A function has an inverse if its graph satisfies the horizontal line test.

ANSWER: TRUE

CONTEST 13 ROUND 1A

Find the range of the given function

1 $y = 4x^2 - 12x + 20$

ANSWER: $\{y : y \geq 11\}$

$[y = 4x^2 - 12x + 9 + 11 = (2x - 3)^2 + 11, \text{ hence } y \geq 11]$

2 $y = 7 - 9x^2 + 12x$

ANSWER: $\{y : y \leq 11\}$

$[y = 7 - (9x^2 - 12x + 4) + 4 = 11 - (3x - 2)^2, \text{ hence } y \leq 11]$

3 $y = 16x^2 + 40x + 20$

ANSWER: $\{y : y \geq -5\}$

$[y = 16x^2 + 40x + 25 - 5 = (4x + 5)^2 - 5, \text{ hence } y \geq -5]$

CONTEST 13 ROUND 1B

A linear transformation is defined by $T: (x, y) \rightarrow (2x + y, 3x - 2y)$. Find the coordinates of the point $P(x, y)$ whose image under T is the point A given

1 $A(1, 5)$ **ANSWER: $P(1, -1)$**

$[2x + y = 1, 3x - 2y = 5, 4x + 2y = 2, 7x = 7, x = 1, y = 1 - 2x = -1, \text{ hence } P(1, -1)]$

2 $A(1, 12)$ **ANSWER: $P(2, -3)$**

$[2x + y = 1, 3x - 2y = 12, 4x + 3y = 14, x = 2, y = 1 - 4 = -3, \text{ hence } P(2, -3)]$

3 $A(-2, -10)$ **ANSWER: $P(-2, 2)$**

$[2x + y = -2, 3x - 2y = -10, 4x + 3y = -4 - 10, x = -2, y = -2x - 2 = 4 - 2 = 2, \text{ hence } P(-2, 2)]$

CONTEST 13 ROUND 2: SPEED RACE

1 Solve for x from the equation $27x^{-3/4} = 64$

ANSWER: $x = 81/256$

[$x^{3/4} = 27/64$, $x^{1/4} = 3/4$, $x = 3^4/4^4 = 81/256$]

2 Find a unit vector perpendicular to the vector **$\mathbf{a} = 12\mathbf{i} - 5\mathbf{j}$**

ANSWER: $(5\mathbf{i} + 12\mathbf{j})/13$

[perpendicular vector is $5\mathbf{i} + 12\mathbf{j}$, unit vector is $(5\mathbf{i} + 12\mathbf{j})/13$]

3 Factorize the polynomial $p(x) = x^4 + 9x^2 - 36$ as a product of 2 binomials.

ANSWER: $(x^2 - 3)(x^2 + 12)$

[$x^4 + 12x^2 - 3x^2 - 36 = x^2(x^2 + 12) - 3(x^2 + 12) = (x^2 - 3)(x^2 + 12)$]

CONTEST 13 ROUND 5: RIDDLE

1 I am the stuff Mathematics is all about.

2 I am a demonstration of a mathematical result or theorem.

3 I am generally a sequence of statements.

4 Each statement follows logically from the preceding statement.

5 The last statement is that which is to be established.

6 I am that which makes a theorem a theorem.

Who am I?

ANSWER: PROOF

CONTEST 13 ROUND 4A: TRUE OR FALSE

1 If y varies directly as x and inversely as z^2 then $y = kx/z^2$

ANSWER: TRUE

2 If y varies inversely as the cube of x , then $y = kx^{-3}$

ANSWER: TRUE

3 If y varies as x , then $xy = k$

ANSWER: FALSE

CONTEST13 ROUND 4B: TRUE OR FALSE

If A , B and C are 2×2 matrices, then

1 $(A + B)(A - B) = A^2 - B^2$

ANSWER: FALSE

2 $(A + B)^2 = A^2 + 2AB + B^2$

ANSWER: FALSE

3 $A + (B + C) = (A + B) + C$

ANSWER: TRUE

CONTEST 14 ROUND 1A

Solve the cubic equation

$$1 \ x^3 - 3x^2 - 25x + 75 = 0$$

ANSWER: $x = 5, -5, 3$

$$[x^2(x - 3) - 25(x - 3) = (x - 5)(x + 5)(x - 3) = 0, x = 5, -5, 3]$$

$$2 \ x^3 - 2x^2 - 16x + 32 = 0$$

ANSWER: $x = 4, -4, 2$

$$[x^2(x - 2) - 16(x - 2) = (x - 2)(x - 4)(x + 4) = 0, x = 4, -4, 2]$$

$$3 \ x^3 - 2x^2 - 15x = 0$$

ANSWER: $x = 5, -3, 0$

$$[x(x^2 - 2x - 15) = x(x - 5)(x + 3) = 0, x = 5, -3, 0]$$

CONTEST 14 ROUND 1B

Differentiate the rational function with respect to x .

$$1 \ y = (2x + 3)/(2x - 1)$$

ANSWER: $dy/dx = -8/(2x - 1)^2$

$$[dy/dx = [2(2x - 1) - 2(2x + 3)]/(2x - 1)^2 = -8/(2x - 1)^2]$$

$$2 \ y = (x + 5)/(2x + 3)$$

ANSWER: $dy/dx = -7/(2x + 3)^2$

$$[dy/dx = [1(2x + 3) - 2(x + 5)]/(2x + 3)^2 = -7/(2x + 3)^2]$$

$$3 \ y = (5x + 2)/(2x - 3)$$

ANSWER: $dy/dx = -19/(2x - 3)^2$

$$[dy/dx = [5(2x - 3) - 2(5x + 2)]/(2x - 3)^2 = -19/(2x - 3)^2]$$

CONTEST 14 ROUND 2: SPEED RACE

1 Three persons are selected from a group of 6 men and 4 women. Find the probability of selecting one man and two women.

ANSWER: $\frac{3}{10}$

$$[(6C_1)(4C_2)/10C_3] = 6(4 \times 3/2)/(10 \times 9 \times 8/3!) = 6 \times 6/120 = 6/20 = 3/10]$$

2 Simplify $\sin(\tan^{-1}(-1))$

ANSWER: $-\sqrt{2}/2$

$$[\tan^{-1}1 = 45^\circ, \sin(-45^\circ) = -\sin 45^\circ = -1/\sqrt{2} = -\sqrt{2}/2]$$

3 Express $\log(x^2 - 36) - \log(x - 6) + \log(x + 6)$ as a single logarithm and simplify.

ANSWER: $2\log(x + 6)$ or $\log(x + 6)^2$

$$[\log(x^2 - 36)/(x - 6) + \log(x + 6) = \log(x + 6) + \log(x + 6) = 2\log(x + 6)]$$

CONTEST 14 ROUND 5: RIDDLE

1 I am a polyhedron.

2 I sit on a polygonal base.

3 I am not a prism.

4 My faces are triangles intersecting in a common vertex.

5 An example of me is a tetrahedron.

6 An extreme example of me is a circular cone.

Who am I?

ANSWER: PYRAMID

CONTEST 14 ROUND 4A: TRUE OR FALSE

1 $(x^2 - 1)/(x + 1) = x - 1$

ANSWER: TRUE

2 $(x^3 - 1)/(x^2 - 1) = (x^2 + x + 1)/(x + 1)$

ANSWER: TRUE

3 $(x^3 + 1)/(x^2 + 1) = (x^2 - x + 1)/(x - 1)$

ANSWER: FALSE

CONTEST 14 ROUND 4B: TRUE OR FALSE

In the following x is a real number

1 $\sqrt{x^2} = x$

ANSWER: FALSE

2 $\sqrt[3]{x^3} = x$

ANSWER: TRUE

3 If $x < 0$, then $\sqrt{x^2} = -x$

ANSWER: TRUE

CONTEST 15 ROUND 1A

Determine the constants a and b given that the polynomial

1 $f(x) = ax^2 + 3x + b$ has $(x - 1)$ as a factor, and leaves a remainder of -9 when divided by $(x - 2)$ **ANSWER: a = -4, b = 1**

$$[a + 3 + b = 0, 4a + 6 + b = -9, 3a = -12, a = -4, -4 + 3 + b = 0, b = 4 - 3 = 1]$$

2 $f(x) = x^2 + ax + b$ has $(x - 2)$ as a factor, and leaves a remainder of 6 when divided by $(x + 1)$. **ANSWER: a = -3, b = 2**

$$[4 + 2a + b = 0, 1 - a + b = 6, b = a + 5, 3a + 9 = 0, a = -3, b = 5 - 3 = 2]$$

3 $f(x) = ax^2 + bx + 5$ has $(x - 1)$ as a factor, and leaves a remainder of 9 on division by $(x + 2)$ **ANSWER: a = -1, b = -4**

$$[a + b + 5 = 0, 4a - 2b + 5 = 9, 6a + 15 = 9, 6a = -6, a = -1 \text{ and } b = -4]$$

CONTEST 15 ROUND 1B

Evaluate

1 $\sin^{-1}(\tan \pi/3)$

ANSWER: NO SOLUTION

$$[\tan(\pi/3) = \sqrt{3}, \sin^{-1}(\sqrt{3}) \text{ does not exist in the domain of } \sin^{-1} \text{ } [-1, 1]]$$

2 $\sin(\tan^{-1}(-1))$

ANSWER: $-\sqrt{2}/2$ or $-1/\sqrt{2}$

$$[\tan^{-1}(-1) = -\pi/4, \sin(-\pi/4) = -\sin(\pi/4) = -1/\sqrt{2} = -\sqrt{2}/2]$$

3 $\cos(2\sin^{-1}(1/2))$

ANSWER: $\frac{1}{2}$

$$[\sin^{-1}(1/2) = \pi/6, \cos(2\sin^{-1}(1/2)) = \cos(\pi/3) = \frac{1}{2}]$$

CONTEST 15 ROUND: SPEED RACE

1 A vector is given in a bearing form as $(10, 120^\circ)$. Express the vector in terms of the unit vectors $\mathbf{i}$ and $\mathbf{j}$

ANSWER: $5\sqrt{3}\mathbf{i} - 5\mathbf{j}$

$$[\mathbf{a} = 10\cos 30^\circ \mathbf{i} - 10\sin 30^\circ \mathbf{j} = 5\sqrt{3}\mathbf{i} - 5\mathbf{j}]$$

2 Simplify the rational expression $(x^2 + 3x - 4)/(x^2 + 2x - 3)$

ANSWER: $(x + 4)/(x + 3)$

$$[(x - 1)(x + 4)/(x - 1)(x + 3) = (x + 4)/(x + 3)]$$

3 Find the coordinates of the point $P(x, y)$ whose image under the linear transformation $T: (x, y) \rightarrow (3x - y, x + y)$ is the point $Q(2, 2)$

ANSWER: $(1, 1)$

$$[3x - y = 2, x + y = 2, 4x = 4, x = 1, y = 2 - x = 2 - 1 = 1, \text{ hence } (1, 1)]$$

CONTEST 15 ROUND 5: RIDDLE

1 I am a 3-digit number.

2 I am a palindromic number.

3 This means forward or backward, the value is the same.

4 All my digits are odd.

5 The sum of my digits is 5.

6 My last digit is an identity.

Who am I?

ANSWER: 131

CONTEST 15 ROUND 4A: TRUE OR FALSE

1 If $\log(\sin x) = 0$, then $\sin x = 0$

ANSWER: FALSE

2 If $\log(\tan x) = 0$, then $\tan x = 1$

ANSWER: TRUE

3 If $\log \cos x = 1$, then $\cos x = 10$

ANSWER: TRUE

CONTEST 15 ROUND 4B: TRUE OR FALSE

In the following a is a real number. The equation $(x - 3)^2 = a$ has

1 real roots,

ANSWER: FALSE

2 rational roots if a is an exact square,

ANSWER: TRUE

3 irrational roots if a is negative.

ANSWER: FALSE

CONTEST 16 ROUND 1A

Find the set of values of k such that the quadratic equation

1 $x^2 + (k + 2)x + k = 0$ has real roots,

ANSWER: $\{k: k \text{ is a real number}\}$ or $\{\text{all real } k\}$

$[(k + 2)^2 - 4k \geq 0, k^2 + 4k + 4 - 4k = k^2 + 4 \geq 0, \text{ valid for all real } k]$

2 $x^2 + (k + 2)x + 2k = 0$ has distinct real roots,

ANSWER: $\{k: k \neq 2\}$

$[(k + 2)^2 - 4(2k) = k^2 - 4k + 4 = (k - 2)^2 > 0, k \neq 2]$

3 $x^2 + 2kx + (k + 2) = 0$ has no real roots.

ANSWER: $\{k: -1 < k < 2\}$

$[4k^2 - 4(k + 2) < 0, k^2 - k - 2 < 0, (k - 2)(k + 1) < 0, -1 < k < 2]$

CONTEST 16 ROUND 1B

Solve the equation

1 $\log(x^{\log x}) = 4$

ANSWER: $x = 100, 1/100$ $[(\log x)(\log x) = 4, \log x = \pm 2, \log x = 2, x = 10^2 = 100, \log x = -2, x = 10^{-2} = 1/100]$

2 $\log(x^3) = (\log x)^2$

ANSWER: $x = 1, 1000$

$[\text{Let } u = \log x, 3u = u^2, u = 0, 3, \log x = 0, x = 1, \log x = 3, x = 10^3 = 1,000]$

3 $\log(x^9) = (\log x)^3$

ANSWER: $x = 1, 1000, 1/1000$ $[\text{Let } u = \log x, 9u = u^3, u(u^2 - 9) = 0, u = 0, \pm 3, \log x = 0, x = 1, \log x = 3, x = 1000, \log x = -3, x = 1/1000]$

CONTEST 16 ROUND 2: SPEED RACE

1 Find the value(s) of x if the matrix $A = \begin{pmatrix} x & x+6 \\ 3 & x \end{pmatrix}$ has no inverse.

ANSWER: $x = 6, -3$

$$[\det A = x^2 - 3(x + 6) = x^2 - 3x - 18 = (x - 6)(x + 3) = 0, x = 6, -3]$$

2 A and B are independent events with $P(A) = \frac{3}{4}$ and $P(B) = \frac{1}{2}$. Find $P(A \cup B)$

ANSWER : $\frac{7}{8}$ $[P(A \cup B) = P(A) + P(B) - P(A)P(B) = \frac{3}{4} + \frac{1}{2} - (\frac{3}{4})(\frac{1}{2}) = \frac{5}{4} - \frac{3}{8} = \frac{7}{8}]$

3 Find the area of an equilateral triangle of sides of length 8 cm.

ANSWER: $16\sqrt{3} \text{ cm}^2$

$$[\text{Area} = \frac{1}{2} (8 \times 8) \sin 60^\circ = 32\sqrt{3}/2 = 16\sqrt{3} \text{ cm}^2]$$

CONTEST 16 ROUND 5: RIDDLE

1 I am a special point on the graph of a function.

2 At me, the tangent changes from negative to positive.

3 Thus I am a turning point.

4 I relate to a least value for the function in my neighborhood.

5 I am characterized by a zero gradient and a positive second derivative.
Who am I?

ANSWER: MINIMUM POINT

CONTEST 16 ROUND 4A: TRUE OR FALSE

1 $1\frac{3}{4} > \sqrt{3}/2$

ANSWER: FALSE

2 $2\frac{4}{5} < 2/\sqrt{5}$

ANSWER: TRUE

3 $\sqrt{7}/2 < 7/4$

ANSWER: TRUE

CONTEST 16 ROUND 4B: TRUE OR FALSE

1 If $\log(x+1) + \log(x-1) = 3$, then $x^2 - 1 = 10^3$

ANSWER: TRUE

2 If $\log_2(x+3) = 2$, then $x+3 = 10^2$

ANSWER: FALSE

3 If $x = (\log y)/k$, then $y = 10^{kx}$

ANSWER: TRUE

CONTEST 17 ROUND 1A

Express as a single logarithm and simplify your answer.

$$1 \log(x^2 - 5x - 14) - \log(x^2 - 4)$$

ANSWER: $\log((x - 7)/(x - 2))$

$$[\log(x - 7)(x + 2)/(x - 2)(x + 2) = \log((x - 7)/(x - 2))]$$

$$2 \log(x^3 - y^3) - \log(x^2 + xy + y^2)$$

ANSWER: $\log(x - y)$

$$[\log((x - y)(x^2 + xy + y^2)/(x^2 + xy + y^2)) = \log(x - y)]$$

$$3 \log(x^4 - y^4) - \log(x^2 + y^2) - \log(x - y)$$

ANSWER: $\log(x + y)$

$$[\log((x^2 + y^2)(x - y)(x + y)/(x^2 + y^2)(x - y)) = \log(x + y)]$$

CONTEST 17 ROUND 1B

4 different Mathematics books and 3 different English books are to be placed on a shelf in a bookshop. Find the number of ways this can be done if

1 the Mathematics books are to be together,

ANSWER: 576

$$[4!(4!) = 24 \times 24 = 576]$$

2 the English books are to be together,

ANSWER: 720

$$[3!(5!) = 6 \times 120 = 720]$$

3 the Mathematics books are to be together and the English books are to be together.

ANSWER: 288

$$[2!(4!)(3!) = 2 \times 24 \times 6 = 288]$$

CONTEST 17 ROUND 2: SPEED RACE

1 Solve the equation $\sin^2 x = \frac{1}{2}$ for the interval $0 < x < \pi$.

ANSWER: $x = \pi/4, 3\pi/4$

[$\sin x = \pm 1/\sqrt{2}$, $\sin x = 1/\sqrt{2}$, $x = \pi/4, 3\pi/4$, $\sin x = -1/\sqrt{2}$ has no solution in the interval]

2 If $i^2 = -1$, find $(1 + i)^2$

ANSWER: $2i$

[$1 + 2i + i^2 = 1 + 2i - 1 = 2i$]

3 Find the inverse of the matrix $A = \begin{pmatrix} 2 & -1 \\ -4 & 2 \end{pmatrix}$

ANSWER: NO SOLUTION, or MATRIX A HAS NO INVERSE

[$\det A = 4 - 4 = 0$, hence matrix has no inverse]

CONTEST 17 ROUND 5: RIDDLE

1 I am a series or a sequence.

2 With me there is the same relation between consecutive terms.

3 I am classified as Arithmetic if there is a constant difference for consecutive terms.

4 I am classified as Geometric if there is a constant ratio for consecutive terms.

5 I am a Progress in Regression.

Who am I?

ANSWER: PROGRESSION

CONTEST 17 ROUND 4A: TRUE OR FALSE

1 $\sin(5\pi/4) = \cos(5\pi/4)$

ANSWER: TRUE

2 $\sin(3\pi/4) = \cos(3\pi/4)$

ANSWER: FALSE

3 $\sin(7\pi/4) = \cos(7\pi/4)$

ANSWER: FALSE

CONTEST 17 ROUND 4B: TRUE OR FALSE

Let a be a positive real number

1 If $a < 1$, then $a < a^2$,

ANSWER: FALSE

2 If $n > m$, then $a^n > a^m$,

ANSWER: FALSE

3 If $a > 1$, then $a^2 > a^3$.

ANSWER: FALSE

CONTEST 18 ROUND 1A

Find values for b, and c such that the quadratic inequality

1 $x^2 + bx + c < 0$ has solution set $\{x: -1 < x < 3\}$

ANSWER: $b = -2, c = -3$

$[(x + 1)(x - 3) < 0, x^2 - 2x - 3 < 0, \text{ hence } b = -2, c = -3]$

2 $x^2 + bx + c > 0$ has solution set $\{x: x > -2 \text{ or } x < -3\}$

ANSWER: $b = 5, c = 6$

$[(x + 2)(x + 3) = x^2 + 5x + 6 > 0, \text{ hence } b = 5, c = 6]$

3 $x^2 + bx + c < 0$ has solution set $\{x: -3 < x < -1\}$

ANSWER: $b = 4, c = 3$

$[(x + 3)(x + 1) = x^2 + 4x + 3 < 0, \text{ hence } b = 4, c = 3]$

CONTEST 18 ROUND 1B

Find the area of the finite region bounded by the given curve and the x-axis.

1 $y = 3x^2 - 6x$

ANSWER: 4

$[3x^2 - 6x = 0, x = 2, 0 \quad A = \int_0^2 (3x^2 - 6x) dx = x^3 - 3x^2 \Big|_0^2 = 8 - 12 = -4, \text{ Area} = |-4| = 4]$

2 $y = 9 - x^2$

ANSWER: 36

$[9 - x^2 = 0, x = \pm 3, A = \int_{-3}^3 (9 - x^2) dx = 9x - x^3 / 3 \Big|_{-3}^3 = 2(27 - 9) = 36]$

3 $y = x^2 - 3x$

ANSWER: 9/2

$[x^2 - 3x = 0, x = 0, 3, A = \int_0^3 (x^2 - 3x) dx = x^3 / 3 - 3x^2 / 2 \Big|_0^3 = (9 - 27/2) = -9/2 \quad \text{Area} = 9/2]$

CONTEST 18 ROUND 2: SPEED RACE

1 Determine the amplitude A and the period T of the function $y = 2\tan(2x)$

ANSWER: A is undefined or ∞ , $T = \pi/2$ or 90° .

[tan has no amplitude, $2T = \pi$, hence $T = \pi/2$]

2 Find the sum of the interior angles of a polygon of 15 sides in degrees.

ANSWER: 2340°

$[(15 - 2)180 = 13 \times 180 = 1800 + 540 = 2340^\circ]$

3 A linear transformation in the plane is defined by T: $(x, y) \rightarrow (2x - y, -x + 2y)$. Find the matrix of the linear transformation.

ANSWER: $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$

CONTEST 18 ROUND 5: RIDDLE

1 I am a trigonometric ratio.

2 I am one of the three basic trigonometric ratios.

3 I take values between -1 and 1 inclusive.

4 This excludes the tangent.

5 I take my maximum value for a zero angle.

6 I am zero for a right angle.

Who am I?

ANSWER: COSINE

CONTEST 18 ROUND 4A: TRUE OR FALSE

In a triangle ABC,

1 If $\sin \angle A = \frac{1}{2}$, then $\angle A = 30^\circ$.

ANSWER: FALSE

2 If $\cos \angle B = -1/2$, then $\angle B = 120^\circ$.

ANSWER: TRUE

3 If $\tan \angle C = -\sqrt{3}$, then $\angle C = 150^\circ$.

ANSWER: FALSE

CONTEST 18 ROUND 4B: TRUE OR FALSE

1 Triangles are similar if corresponding angles are congruent.

ANSWER: TRUE

2 Hexagons are similar if corresponding angles are congruent.

ANSWER: FALSE

3 Nonagons are similar if corresponding angles are congruent.

ANSWER: TRUE

CONTEST 19 ROUND 1A

Solve the compound inequality for x.

1 $x^2 < 4$, or $x^2 > 9$

ANSWER: $\{x: x < -3, \text{ or } -2 < x < 2, \text{ or } x > 3\}$

$[\{x: -2 < x < 2\} \cup \{x: x > 3 \text{ or } x < -3\}] = \{x: x < -3, \text{ or } -2 < x < 2, \text{ or } x > 3\}$

2 $x^2 < 1$ and $x^2 > 16$

ANSWER: empty set or null set

$[\{x: -1 < x < 1\} \cap \{x: x < -4 \text{ or } x > 4\}] = \{ \}$

3 $x^2 < 25$ and $x^2 > 4$

ANSWER: $\{x: -5 < x < -2 \text{ or } 2 < x < 5\}$

$[\{x: -5 < x < 5\} \cap \{x: x < -2 \text{ or } x > 2\}] = \{x: -5 < x < -2 \text{ or } 2 < x < 5\}$

CONTEST 19 ROUND 1B

Find a cartesian equation of the curve given by the pair of equations

1 $x = 2t$, $y = 2/t$

ANSWER: $xy = 4$, or $y = 4/x$

$[xy = 2t(2/t) = 4]$

2 $x = 2t^2$, $y = 2t$

ANSWER: $y^2 = 2x$

$[y^2 = 4t^2 = 2(2t^2) = 2x, y^2 = 2x]$

3 $x = t - 1$, $y = 5 - 3t$

ANSWER: $y = 2 - 3x$, or $3x + y = 2$

$[t = x + 1, y = 5 - 3(x + 1) = 2 - 3x]$

CONTEST 19 ROUND 2: SPEED RACE

1 Differentiate the function $y = \sqrt{(2x^3 - 4x)}$ with respect to x .

ANSWER: $\frac{dy}{dx} = (3x^2 - 2)(2x^3 - 4x)^{-1/2}$

$$[dy/dx = \frac{1}{2} (6x^2 - 4)(2x^3 - 4x)^{-1/2} = (3x^2 - 2)(2x^3 - 4x)^{-1/2}]$$

2 Find the length of the altitude of a triangle if its base is of length 8 cm and its area is equal to area of a square of diagonal of length 4 cm.

ANSWER: 2 cm $[(1/2) \times 8h = \frac{1}{2} (16) = 8, 4h = 8, h = 2]$

3 Two events A and B are disjoint and $P(A) = 1/3$ and $P(A \cup B) = 3/4$. Find $P(B)$.

ANSWER: 5/12

$$[P(A \cup B) = P(A) + P(B), P(B) = P(A \cup B) - P(A) = (3/4) - (1/3) = 5/12]$$

CONTEST 19 ROUND 5: RIDDLE

1 I am a 3-dimensional figure.

2 I am formed of faces.

3 The faces intersect in edges.

4 The edges intersect in points called vertices.

5 In my regular form, my faces are congruent and my edges are congruent.

6 Examples of me are the cube, the tetrahedron and the octahedron.

Who am I?

ANSWER: POLYHEDRON

CONTEST 19 ROUND 4A: TRUE OR FALSE

Given that $0 < x < \pi$,

1 If $\cos x = \frac{1}{2}$, then $\sin x = \frac{\sqrt{3}}{2}$

ANSWER: TRUE

2 If $\cos x = -\frac{\sqrt{3}}{2}$ then $\sin x = -\frac{1}{2}$

ANSWER: FALSE

3 If $\tan x = 1$, then $\sin x = \frac{1}{\sqrt{2}}$

ANSWER: TRUE

CONTEST 19 ROUND 4B: TRUE OR FALSE

1 The line through the points A(2, 3) and B(2, 5) has an undefined slope

ANSWER: TRUE

2 The line through the points A(-2, 7) and B(2, 7) has an undefined slope.

ANSWER: FALSE

3 The slope of the line intersecting the y-axis at 6 and the x-axis at 3 is -2.

ANSWER: TRUE

CONTEST 20 ROUND 1A

Find the length of the side of an equilateral triangle if its area is

1 $36\sqrt{3} \text{ cm}^2$

ANSWER: 12 cm

$[(1/2)a^2\sin 60^\circ = (\frac{1}{2})a^2 \sqrt{3}/2 = 36\sqrt{3} \text{ cm}^2, a^2 = 4 \times 36, a = 12]$

2 $81\sqrt{3} \text{ cm}^2$

ANSWER: 18 cm

$[(1/2)a^2\sin 60^\circ = (\frac{1}{2})a^2 \sqrt{3}/2 = 81\sqrt{3}, a^2 = 4 \times 81, a = 2 \times 9 = 18 \text{ cm}]$

3 $121\sqrt{3} \text{ cm}^2$

ANSWER: 22 cm

$[(1/2)a^2\sin 60^\circ = (\frac{1}{2})a^2 \sqrt{3}/2 = 121\sqrt{3}, a^2 = 4 \times 121, a = 2 \times 11 = 22 \text{ cm}]$

CONTEST 20 ROUND 1B

Determine the value of k so that the line with equation

1 $ky = 4x + 6$ has y-intercept -2

ANSWER: k = -3

[Let $x = 0, k(-2) = 6, k = -3$]

2 $y = kx + 5$ has x-intercept -1

ANSWER: k = 5

[Let $y = 0, 0 = (-1)k + 5, k = 5$]

3 $ky = 3x + 5$ has slope -2

ANSWER: k = -3/2

$[y = (3/k)x + 5/k, 3/k = -2, k = -3/2]$

CONTEST 20 ROUND 2: SPEED RACE

1 A function f is defined by $f(a + b) = f(a) + f(b)$ and $f(1) = 2$. Evaluate $f(3)$.

ANSWER: 6

$[f(2) = f(1) + f(1) = 2 + 2 = 4, f(3) = f(2) + f(1) = 4 + 2 = 6]$

2 Solve for x in terms of b and d from the equation $x^2 - 3dx + 5bx - 15bd = 0$

ANSWER: $x = 3d, -5b$

$[x(x - 3d) + 5b(x - 3d) = (x - 3d)(x + 5b) = 0, x = 3d, -5b]$

3 Find constants A and B such that $2x/(x^2 - 1) = A/(x - 1) + B/(x + 1)$

ANSWER: $A = 1, B = 1$

$[2x = A(x + 1) + B(x - 1), \text{ let } x = 1, 2 = 2A, A = 1, \text{ let } x = -1, -2 = -2B, B = 1]$

CONTEST 20 ROUND 5: RIDDLE

1 I am an angle.

2 To be precise I am a type of angle.

3 I am common in triangles.

4 Every triangle has at least two of me.

5 My sine or cosine is non-negative.

6 I have a measure less than that of a right angle.

Who am I?

ANSWER: ACUTE ANGLE

CONTEST 20 ROUND 4A: TRUE OR FALSE

1 $nC_r = n!/(n-r)!r!$ (Read as n combination r)

ANSWER: TRUE

2 $nP_r = n!/r!$ (Read as n permutation r)

ANSWER: FALSE

3 $nC_r = r!(nP_r)$

ANSWER: FALSE

CONTEST 20 ROUND 4B: TRUE OR FALSE

1 $\sin^{-1}(-1/\sqrt{2}) = 3\pi/4$

ANSWER: FALSE

2 $\cos^{-1}(-1/\sqrt{2}) = 3\pi/4$

ANSWER: TRUE

3 $\tan^{-1}(-1) = 3\pi/4$

ANSWER: FALSE

CONTEST 21 ROUND 1A

Find the cosine of the angle between the given vectors **a** and **b**.

1 **a** = **3i + 5j** and **b** = **5i - 3j**

ANSWER: 0

$$[\cos\theta = \mathbf{a} \cdot \mathbf{b} / |\mathbf{a}| |\mathbf{b}| = (15 - 15) / 34 = 0]$$

2 **a** = **4i - 3j** and **b** = **i - 2j**

ANSWER: $2/\sqrt{5}$ or $2\sqrt{5}/5$

$$[\cos\theta = \mathbf{a} \cdot \mathbf{b} / |\mathbf{a}| |\mathbf{b}| = (4 + 6) / 5\sqrt{5} = 10 / 5\sqrt{5} = 2/\sqrt{5}]$$

3 **a** = **-2i + 3j** and **b** = **2i + j**

ANSWER: $-1/\sqrt{65}$ or $-\sqrt{65}/65$

$$[\cos\theta = \mathbf{a} \cdot \mathbf{b} / |\mathbf{a}| |\mathbf{b}| = (-4 + 3) / \sqrt{13}\sqrt{5} = -1/\sqrt{65}]$$

CONTEST 21 ROUND 1B

Find the coordinates of the stationary points of the given curve

1 $y = x^2(3 - 2x)$

ANSWER: (0, 0), (1, 1)

[$y = 3x^2 - 2x^3$, $dy/dx = 6x - 6x^2 = 6x(1 - x) = 0$, $x = 0, 1$. For $x = 1$, $y = 1$, and for $x = 0$, $y = 0$, hence (0, 0), (1, 1)]

2 $y = x^3(4 + 3x)$

ANSWER: (0, 0), (-1, -1)

[$y = 4x^3 + 3x^4$, $dy/dx = 12x^2 + 12x^3 = 0$, $x = 0, -1$. For $x = 0$, $y = 0$, and for $x = -1$, $y = -1$, hence (0, 0) and (-1, -1)]

3 $y = x^4(4x - 5)$

ANSWER: (0, 0), (1, -1)

[$y = 4x^5 - 5x^4$, $dy/dx = 20x^4 - 20x^3 = 20x^3(x - 1) = 0$, $x = 0, 1$
For $x = 0$, $y = 0$ and for $x = 1$, $y = -1$, hence (0, 0) and (1, -1)]

CONTEST 21 ROUND 2: SPEED RACE

1 Find an equivalent but simpler trigonometric expression for

$$y = \operatorname{cosec}(x + \pi/2)$$

ANSWER: $y = \sec x$

$$[\sin(x + \pi/2) = \cos x, \text{ hence } \operatorname{cosec}(x + \pi/2) = 1/\cos x = \sec x]$$

2 Solve the equation $\log(2\sin x) = 0$, for x in the interval $0 \leq x \leq \pi/2$.

ANSWER: $x = \pi/6$

$$[2\sin x = 1, \sin x = 1/2, x = \pi/6]$$

3 Find the domain of the function $f(x) = \sqrt[3]{(3x + 2)}$

ANSWER: $\{x: x \text{ is a real number}\}$ or $\{\text{real numbers}\}$

CONTEST 21 ROUND 5: RIDDLE

1 I am a scalar quantity.

2 I am in a way a function of a square matrix.

3 I am a sum or difference of products of terms of a square matrix.

4 For a 2×2 matrix, I am the difference of the products of the diagonal terms.

5 I am to a square matrix, what a discriminant is to a quadratic equation.

6 I determine if a square matrix has an inverse or not and hence my name.

Who am I?

ANSWER: DETERMINANT

CONTEST 21 ROUND 4A: TRUE OR FALSE

1 If a number is not divisible by 9, it is not divisible by 3.

ANSWER: FALSE

2 If a number is not divisible by 3, it is not divisible by 9.

ANSWER: TRUE

3 If a number is not divisible by 15, it is not divisible by 5.

ANSWER: FALSE

CONTEST 21 ROUND 4B: TRUE OR FALSE

A quadratic curve has equation $y = 2(x + 3)^2 - 5$.

1 The vertex is (3, -5).

ANSWER: FALSE

2 The axis of symmetry is $x = -3$.

ANSWER: TRUE

3 the minimum value of y is -5.

ANSWER: TRUE

CONTEST 22 ROUND 1A

Find the matrix product if possible. If not possible, give a reason.

$$1 \begin{pmatrix} 1 & -1 \\ 2 & 0 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ 1 & -1 \end{pmatrix}$$

$$\text{ANSWER: } \begin{pmatrix} 1 & 0 \\ 4 & -2 \\ 0 & -1 \end{pmatrix}$$

$$2 \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} -1 & 2 & 1 \\ 2 & 0 & -1 \end{pmatrix}$$

$$\text{ANSWER: } \begin{pmatrix} -4 & 4 & 3 \\ 3 & 2 & -1 \end{pmatrix}$$

$$3 \begin{pmatrix} 1 & -2 & -1 \\ 2 & 3 & 1 \end{pmatrix} \begin{pmatrix} -2 & 1 & -2 \\ 1 & -1 & 2 \end{pmatrix}$$

ANSWER: NOT POSSIBLE since number of columns of first matrix is not equal to the number of rows of second matrix $(2 \neq 2 + 1)$

CONTEST 22 ROUND 1B

Factorise completely

$$1 \ 24x^3 - 375y^3$$

$$\text{ANSWER: } 3(2x - 5y)(4x^2 + 10xy + 25y^2)$$

$$[3((2x)^3 - (5y)^3) = 3(2x - 5y)(4x^2 + 10xy + 25y^2)]$$

$$2 \ 54x^3 + 16y^3$$

$$\text{ANSWER: } 2(3x + 2y)(9x^2 - 6xy + 4y^2)$$

$$[2(27x^3 + 8y^3) = 2(3x + 2y)(9x^2 - 6xy + 4y^2)]$$

$$3 \ 500x^3 - 108y^3$$

$$\text{ANSWER: } 4(5x - 3y)(25x^2 + 15xy + 9y^2)$$

$$[4(125x^3 - 27y^3) = 4(5x - 3y)(25x^2 + 15xy + 9y^2)]$$

CONTEST 22 ROUND 2: SPEED RACE

1 Find the area of the finite region bounded by the curves $y = x^3$ and $y = x^2$.

ANSWER: 1/12

$[x^3 = x^2, x^2(x - 1) = 0, x = 0, 1, \text{Area is given by}]$

$$\left[\int_0^1 (x^2 - x^3) dx = x^3/3 - x^4/4 \right]_0^1, 1/3 - 1/4 = 1/12 = 1/12]$$

2 Express as an exponential $\sqrt[3]{7}$

ANSWER $7^{1/6}$

$$[(7^{1/3})^{1/2} = 7^{1/2 \times 1/3} = 7^{1/6}]$$

3 Solve for x from the absolute value equation $|x^2 - 4x + 4| = 9$

ANSWER: $x = 5, -1$

$$[(x - 2)^2 = 9, x - 2 = \pm 3, x = 5, -1]$$

CONTEST 22 ROUND 5: RIDDLE

1 We are a pair of numbers.

2 We have opposite signs.

3 We are odd numbers.

4 Our sum is -6.

5 Our product is -7.

Who are we?

ANSWER: 1 and -7

CONTEST 22 ROUND 4A: TRUE OR FALSE

1 The logarithm of a number depends only on the number.

ANSWER: FALSE

2 The logarithm of a number depends on the number as well as the base.

ANSWER: TRUE

3 The base of a logarithm can be any real number.

ANSWER: FALSE

CONTEST 22 ROUND 4B: TRUE OR FALSE

1 The sum of two irrational numbers is an irrational number

ANSWER: FALSE

2 The product of two irrational numbers may be a rational number

ANSWER: TRUE

3 The square root of a prime number is an irrational number

ANSWER: TRUE

CONTEST 23 ROUND 1A

Simplify the sum

1 $3\sqrt{128} - 5\sqrt{32} + 2\sqrt{50}$

ANSWER: $14\sqrt{2}$

$[3(8)\sqrt{2} - 5(4)\sqrt{2} + 2(5)\sqrt{2} = (24 - 20 + 10)\sqrt{2} = 14\sqrt{2}]$

2 $6\sqrt{63} + 5\sqrt{175} - 10\sqrt{112}$

ANSWER: $3\sqrt{7}$

$[6(3)\sqrt{7} + 5(5)\sqrt{7} - 10(4)\sqrt{7} = (18 + 25 - 40)\sqrt{7} = 3\sqrt{7}]$

3 $5\sqrt{147} - 7\sqrt{108} + 2\sqrt{243}$

ANSWER: $11\sqrt{3}$

$[5(7)\sqrt{3} - 7(6)\sqrt{3} + 2(9)\sqrt{3} = (35 - 42 + 18)\sqrt{3} = 11\sqrt{3}]$

CONTEST 23 ROUND 1B

Express y in terms of x given that

1 the vectors $\mathbf{a} = (x + 2)\mathbf{i} - 3\mathbf{j}$ and $\mathbf{b} = 5\mathbf{i} + y\mathbf{j}$ are perpendicular,

ANSWER: $y = 5x/3 + 10/3$

$[\mathbf{a} \cdot \mathbf{b} = 5(x + 2) - 3y = 0, y = 5x/3 + 10/3]$

2 the vectors $\mathbf{a} = 5\mathbf{i} + y\mathbf{j}$ and $\mathbf{b} = x\mathbf{i} + 5\mathbf{j}$ are parallel,

ANSWER: $y = 25/x$

$[y/5 = 5/x, xy = 25, y = 25/x]$

3 the vectors $\mathbf{a} = 9x\mathbf{i} + y\mathbf{j}$ and $\mathbf{b} = x\mathbf{i} - y\mathbf{j}$ are perpendicular.

ANSWER: $y = \pm 3x$

$[\mathbf{a} \cdot \mathbf{b} = 9x^2 - y^2 = 0, y^2 = 9x^2, y = \pm 3x]$

CONTEST 23 ROUND 2: SPEED RACE

1 Find the first three terms of the sequence U_n given by $U_n = (\sin(\pi/3))^n$

ANSWER: $U_1 = \sqrt{3}/2, U_2 = 3/4, U_3 = 3\sqrt{3}/8$

$$[U_1 = \sin(\pi/3) = \sqrt{3}/2, U_2 = (\sqrt{3}/2)^2 = 3/4, U_3 = (\sqrt{3}/2)^3 = 3\sqrt{3}/8]$$

2 Find the sum to infinity of the series $4 + 4/3 + 4/9 + 4/27 + \dots$

ANSWER: 6

$$[GP, a = 4, r = 1/3, S_\infty = a/(1 - r) = 4/(1 - 1/3) = 4/(2/3) = 6]$$

3 Given that $x = 4 + \sqrt{17}$ evaluate $(x - 1/x)$ and simplify your answer.

ANSWER: 8

$$[4 + \sqrt{17} - 1/(4 + \sqrt{17}) = 4 + \sqrt{17} - (4 - \sqrt{17})/(-1) = 8]$$

CONTEST 23 ROUND 5: RIDDLE

1 I am a statistical diagram.

2 I am a plot of two sets of measurements.

3 I give an indication about the correlation between the two sets of measurements.

4 I consist of points scattered all over the plane.

5 That accounts for my name.

Who am I?

ANSWER: SCATTER DIAGRAM

CONTEST 23 ROUND 4A: TRUE OR FALSE

1 If the product of two real numbers is a rational number, then both numbers are rational.

ANSWER: FALSE

2 If the product of two real numbers is an irrational number, then both numbers are irrational.

ANSWER: FALSE

3 If the sum of two real numbers is a rational number, then both numbers are rational numbers.

ANSWER: FALSE

CONTEST 23 ROUND 4B: TRUE OR FALSE

If the diagonals of a quadrilateral bisect each other perpendicularly, then it is

1 a rectangle,

ANSWER: FALSE

2 a square,

ANSWER: FALSE

3 a rhombus.

ANSWER: TRUE

CONTEST 24 ROUND 1A

In a rhombus, find

1 lengths of the diagonals if the lengths of the diagonals are in the ratio 3 : 2 and the area is 75 cm^2 , **ANSWER: 15 cm, 10 cm**
[Area = $\frac{1}{2} (3x)(2x) = 3x^2 = 75$, $x^2 = 25$, $x = 5$, $3x = 15$, $2x = 10$]

2 the lengths of the diagonals if one diagonal is thrice as long as the other and the area is 96 cm^2 , **ANSWER: 24 cm, 8 cm**
[Area = $\frac{1}{2} (3x)x = 3x^2/2 = 96$, $x^2 = 64$, $x = 8$, $3x = 24$]

3 the length of a side of the rhombus if its area is 216 cm^2 and a diagonal is of length 18 cm. **ANSWER: 15 cm**
[(1/2)d x 18 = 216, $9d = 216$, $d = 24$, length of side, $a^2 = 12^2 + 9^2$, $a = 15$]

CONTEST 24 ROUND 1B

Find the sum to infinity of the given series. Leave answer as a rational number in simplest form.

1 $10 + 1.0 + 0.1 + 0.01 + 0.001 \dots$

ANSWER: 100/9

[GP, $a = 10$, $r = 0.1$, $S_{\infty} = a/(1 - r) = 10/(1 - 0.1) = 10/0.9 = 100/9$]

2 $10 - 5 + 5/2 - 5/4 + 5/8 - 5/16 + \dots$

ANSWER: 20/3

[GP, $a = 10$, $r = -1/2$, $S_{\infty} = a/(1 - r) = 10/(1 + 1/2) = 10/(3/2) = 20/3$]

3 $63 - 21 + 7 - 7/3 + 7/9 - \dots$

ANSWER : 189/4

[GP, $a = 63$, $r = -1/3$, $S_{\infty} = a/(1 - r) = 63/(1 + 1/3) = 63/(4/3) = 189/4$]

CONTEST 24 ROUND 2: SPEED RACE

1 Find the coordinates of the stationary points of the curve $y = 2x^3 - 6x + 5$.

ANSWER: (1, 1) and (-1, 9)

$[dy/dx = 6x^2 - 6 = 6(x - 1)(x + 1) = 0, x = \pm 1$. For $x = 1, y = 2 - 6 + 5 = 1$, for $x = -1, y = -2 + 6 + 5 = 9]$

2 The equation $ax^2 - bx + c = 0$ has roots α, β . Express $(\alpha - \beta)^2$ in terms of a, b and c

ANSWER: $(b^2 - 4ac)/a^2$

$[(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta = (b/a)^2 - 4(c/a) = (b^2 - 4ac)/a^2]$

3 Simplify the trigonometric expression $(\sin x \cos^3 x + \sin^3 x \cos x)/\sin 2x$
(Assume $\sin 2x \neq 0$)

ANSWER: $1/2$

$[\sin x \cos x (\cos^2 x + \sin^2 x)/2 \sin x \cos x = \sin x \cos x / 2 \sin x \cos x = 1/2]$

CONTEST 24 ROUND 5: RIDDLE

1 I have a magnitude and a direction.

2 That makes me a vector quantity.

3 I am an influence.

4 I influence motion especially in the matter of acceleration.

5 To be precise I am the rate of change of momentum.

6 In simple terms, I am a push or a pull.

Who am I?

ANSWER: FORCE

CONTEST 24 ROUND 4A: TRUE OR FALSE

x and y are real numbers.

1 If $y^2 = x^3$, then x is non-negative

ANSWER: TRUE

2 If $y^2 = x^4$, then $y = x^2$

ANSWER: FALSE

3 If $y^3 = x^2$, then $y \geq 0$ for any x.

ANSWER: TRUE

CONTEST 24 ROUND 4B: TRUE OR FALSE

1 The equation $x^2 - 5x - 6 = 0$ has roots 6 and -1.

ANSWER: TRUE

2 The equation $x^2 + 5x - 14 = 0$ has roots 7 and -2.

ANSWER: FALSE

3 The equation $x^2 + 7x - 10 = 0$ has roots -2 and -5.

ANSWER: FALSE

CONTEST 25 ROUND 1A

Write an inequality which defines the given region in the cartesian x-y plane

1 inside the circle with center (2, -1) and radius 6,

ANSWER: $(x - 2)^2 + (y + 1)^2 < 36$

[circle is $(x - 2)^2 + (y + 1)^2 = 36$, hence inside gives $(x - 2)^2 + (y + 1)^2 < 36$]

2 outside the circle with center (3, 2) and radius 5,

ANSWER: $(x - 3)^2 + (y - 2)^2 > 25$

[circle is $(x - 3)^2 + (y - 2)^2 = 25$, so outside circle gives $(x - 3)^2 + (y - 2)^2 > 25$]

3 inside the circle with center (-3, 5) and with radius 7.

ANSWER: $(x + 3)^2 + (y - 5)^2 < 49$

[circle is $(x + 3)^2 + (y - 5)^2 = 49$, hence inside circle is $(x + 3)^2 + (y - 5)^2 < 49$]

CONTEST 25 ROUND 1B

A and B are events in a sample space S. Find

1 $P(A \cup B)$ if A and B are independent and $P(A) = \frac{3}{4}$, $P(B) = \frac{1}{3}$

ANSWER: $\frac{5}{6}$ [$P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{3}{4} + \frac{1}{3} - (\frac{3}{4})(\frac{1}{3}) = (\frac{3}{4} + \frac{1}{3} - \frac{1}{4}) = \frac{1}{2} + \frac{1}{3} = \frac{5}{6}$]

2 $P(A \cup B)$ given $P(A) = \frac{3}{5}$, $P(B) = \frac{2}{3}$ and $P(A \cap B) = \frac{1}{3}$,

ANSWER: $\frac{14}{15}$ [$P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{3}{5} + \frac{2}{3} - \frac{1}{3} = \frac{3}{5} + \frac{1}{3} = \frac{(9 + 5)}{15} = \frac{14}{15}$]

3 $P(A \cap B)$ given $P(A) = \frac{4}{7}$, $P(B) = \frac{1}{2}$ and $P(A \cup B) = \frac{3}{4}$

ANSWER: $\frac{9}{28}$ [$P(A \cap B) = P(A) + P(B) - P(A \cup B) = \frac{4}{7} + \frac{1}{2} - \frac{3}{4} = \frac{(16 + 14 - 21)}{28} = \frac{9}{28}$]

CONTEST 25 ROUND 2: SPEED RACE

1 Find the area of the finite region bounded by the curves $y = x^2 - 4$ and $y = 0$.

ANSWER: 32/3 sq. units

$$[x^2 - 4 = 0, x = \pm 2, A = \int_{-2}^2 (4 - x^2) dx = 4x - x^3 / 3]_{-2}^2 = (8 - 8/3) - (-8 + 8/3) = 16/3 + 16/3 = 32/3]$$

2 Evaluate and simplify $\cos(\sin^{-1} 1/\sqrt{2})$ **ANSWER: $\sqrt{2}/2$**

$$[\sin^{-1} 1/\sqrt{2} = \pi/4, \cos(\pi/4) = 1/\sqrt{2} = \sqrt{2}/2]$$

3 Find the matrix sum

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & -1 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 2 \\ 0 & 2 & 1 \end{pmatrix} \quad \textbf{ANSWER:} \quad \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

CONTEST 25 ROUND 5: RIDDLE

1 I am an equation

2 To be precise I am a quadratic equation.

3 I definitely have real roots.

4 The sum of my roots is -7.

5 The product of my roots is one less the sum of my roots.

Who am I?

ANSWER: $x^2 + 7x - 8 = 0$

CONTEST 25 ROUND 4A: TRUE OR FALSE

If the interior angle of a regular polygon is

1 120° , then it is a hexagon.

ANSWER: TRUE

2 150° , then it is a duo-decagon.

ANSWER: TRUE

3 140° , then it is an octagon.

ANSWER: FALSE

CONTEST 25 ROUND 4B: TRUE OR FALSE

1 The equation $\sqrt{x - 4} = 1$ has $x = 3$ as a solution.

ANSWER: FALSE

2 The equation $\sqrt{x - 3} = -2$ has $x = 7$ as a solution.

ANSWER: FALSE

3 The equation $\sqrt{x + 5} = 3$ has $x = 4$ as a solution.

ANSWER: TRUE

CONTEST 26 ROUND 1A

Find a unit vector perpendicular to the vector **a**.

1 **a** = $2\mathbf{i} + 3\mathbf{j}$

ANSWER: $(3\mathbf{i} - 2\mathbf{j})/\sqrt{13}$

[**b** = $3\mathbf{i} - 2\mathbf{j}$ is perpendicular to **a**, unit vector is $(3\mathbf{i} - 2\mathbf{j})/\sqrt{13}$]

2 **a** = $5\mathbf{i} - 4\mathbf{j}$

ANSWER: $(4\mathbf{i} + 5\mathbf{j})/\sqrt{41}$

[**b** = $4\mathbf{i} + 5\mathbf{j}$ is perpendicular to **a**, unit vector is $(4\mathbf{i} + 5\mathbf{j})/\sqrt{41}$]

3 **a** = $6\mathbf{i} + 8\mathbf{j}$

ANSWER: $(4\mathbf{i} - 3\mathbf{j})/5$

[**b** = $8\mathbf{i} - 6\mathbf{j}$ is perpendicular to **a**, unit vector is $(8\mathbf{i} - 6\mathbf{j})/10 = (4\mathbf{i} - 3\mathbf{j})/5$]

CONTEST 26 ROUND 1B

Consider the equation $y = A\sin(Bx - C) + D$ in relation to $y = \sin x$. Determine the constant associated with

1 vertical translation,

ANSWER: D

2 horizontal translation,

ANSWER: C

3 vertical stretch or shrink.

ANSWER: A

CONTEST 26 ROUND 2: SPEED RACE

1 Convert the decimal number 486 to a number in base 8.

ANSWER: 746_8

[Highest power of 8 in 486 is $8^2 = 64$, $486 = 7(64) + 38 = 7(64) + 4(8) + 6(1)$, hence $486 = 746_8$]

2 Solve the quadratic equation $3x^2 - 4x - 2 = 0$ and leave answer as a surd.

ANSWER: $x = (2 \pm \sqrt{10})/3$

[$x = (4 \pm \sqrt{(16 + 24)})/2(3) = (4 \pm \sqrt{40})/6 = (2 \pm \sqrt{10})/3$]

3 A binary operation $*$ is defined on the set of real numbers by

$a*b = a\sqrt{2} + b\sqrt{3}$. Evaluate $(2 * 3)*5$

ANSWER: $4 + 5\sqrt{3} + 3\sqrt{6}$

[$2 * 3 = 2\sqrt{2} + 3\sqrt{3}$, $(2\sqrt{2} + 3\sqrt{3})*5 = (2\sqrt{2} + 3\sqrt{3})*\sqrt{2} + 5\sqrt{3} = 4 + 5\sqrt{3} + 3\sqrt{6}$]

CONTEST 26 ROUND 5: RIDDLE

1 I am a binary operation on the set of real numbers.

2 I involve splitting into parts.

3 I am neither commutative nor associative.

4 I am the act of separating something into parts.

5 I give rise to a quotient and a remainder.

Who am I?

ANSWER: DIVISION

CONTEST 26 ROUND 4A: TRUE OR FALSE

$$1 \frac{3}{4} < \sqrt{3}/2$$

ANSWER: TRUE

$$2 \sqrt{5}/2 < 5/4$$

ANSWER: TRUE

$$3 \frac{9}{16} < 3/4$$

ANSWER: TRUE

CONTEST 26 ROUND 4B: TRUE OR FALSE

P and Q are simple statements. The compound statement 'if P then Q' is true whenever

1 P is true

ANSWER: FALSE

2 Q is false

ANSWER: FALSE

3 P is false

ANSWER: TRUE

CONTEST 27 ROUND 1A

Determine the maximum or minimum value of the given quadratic function.

1 $f(x) = -x^2 + 6x - 11$

ANSWER: maximum value of - 2 $[f(x) = -(x^2 - 6x) - 11 =$
 $-(x^2 - 6x + 9) - 11 + 9 = -(x - 3)^2 - 2, \text{ hence a maximum value of } - 2]$

2 $g(x) = 3x^2 - 12x + 16$

ANSWER: minimum value of 4 $[g(x) = 3(x^2 - 4x) + 16 =$
 $3(x^2 - 4x + 4) + 16 - 12 = 3(x - 2)^2 + 4, \text{ hence a minimum value of } 4]$

3 $h(x) = -2x^2 - 12x - 15$

ANSWER: maximum value of 3 $[h(x) = -2(x^2 + 6x) - 15 =$
 $-2(x^2 + 6x + 9) + 18 - 15 = -2(x + 3)^2 + 3, \text{ hence a maximum value of } 3]$

CONTEST 27 ROUND 1B

Solve the equation for x. You may leave answer as an exponential

1 $\log_2(\log_3 x) = 1$

ANSWER: $x = 3^2$, or $x = 9$
 $[\log_3 x = 2, x = 3^2 = 9]$

2 $\log_3(\log_2 x) = 2$

ANSWER: $x = 2^9$
 $[\log_2 x = 3^2 = 9, x = 2^9]$

3 $\log_5(\log_4 x) = 3$

ANSWER: $x = 4^{125}$
 $[\log_4 x = 5^3 = 125, x = 4^{125}]$

CONTEST 27 ROUND 2: SPEED RACE

1 A binary operation is defined on the set Z of integers by $a * b = a^2 - ab + b^2$.
Solve for x from the equation $x * 3 = 37$

ANSWER: $x = 7, -4$

$$[x^2 - 3x + 9 = 37, x^2 - 3x - 28 = (x - 7)(x + 4) = 0, x = 7, -4]$$

2 Find the equation of a curve with gradient $dy/dx = 3x^2 - 2x + 1$ and passing through the origin.

ANSWER: $y = x^3 - x^2 + x$

$$[y = x^3 - x^2 + x + C, \text{ origin gives } 0 = 0 - 0 + 0 + C, C = 0, \text{ hence } y = x^3 - x^2 + x]$$

3 Find the solution set of the absolute value inequality $|2x - 5| < 9$

ANSWER: $\{x: -2 < x < 7\}$

$$[-9 < 2x - 5 < 9, -4 < 2x < 14, -2 < x < 7]$$

CONTEST 27 ROUND 5: RIDDLE

1 I am a compound statement formed from two simple statements.

2 I am true in 3 out of 4 instances.

3 I am characterized by the connective 'or'.

4 I am always true except when both simple statements are false.

5 I take the form 'P or Q'.

6 I am a part of a Disc and a part of an Injunction.

Who am I?

ANSWER: DISJUNCTION

CONTEST 27 ROUND 4A: TRUE OR FALSE

1 In a kite, the diagonals are perpendicular and bisect each other.

ANSWER: FALSE

2 In a rhombus, the diagonals bisect each other perpendicularly and are congruent.

ANSWER: FALSE

3 In a rectangle, the diagonals bisect each other perpendicularly and are congruent.

ANSWER: FALSE

CONTEST 27 ROUND 4B: TRUE OR FALSE

1 $(x - 2)$ is a factor of $f(x) = 3x^3 - 5x^2 - 2x$

ANSWER: TRUE

2 $(x + 2)$ is a factor of $2x^3 + 7x^2 + 3x + 1$

ANSWER: FALSE

3 $(x - 1)$ is a factor of $f(x) = 2x^3 - 9x^2 + 5x + 2$

ANSWER: TRUE

CONTEST 28 (Q/F) ROUND 1A

Find the area of the finite region bounded by the given line and the quadratic curve if

1 the line is $y = 2x$ and the quadratic curve is $y = x^2$,

ANSWER: 4/3

$$[\text{Limits, } x^2 = 2x, x = 0, 2 \quad A = \int_0^2 (2x - x^2) dx = x^2 - x^3 / 3 \Big|_0^2 = 4 - 8/3 = 4/3]$$

2 the line is $y = x$, and the quadratic curve is $y = 2x - x^2$

ANSWER: 1/6 [Limits, $2x - x^2 = x, x^2 - x = 0, x = 0, 1$

$$A = \int_0^1 (2x - x^2 - x) dx = \int_0^1 (x - x^2) dx = x^2 / 2 - x^3 / 3 \Big|_0^1 = 1/2 - 1/3 = 1/6]$$

3 the line is $y = 3x$ and the curve is $y = 2x^2 + x$

ANSWER: 1/3 [Limits, $2x^2 + x = 3x, 2x^2 = 2x, x = 0, 1$

$$A = \int_0^1 (3x - 2x^2 - x) dx = \int_0^1 (2x - 2x^2) dx = x^2 - 2x^3 / 3 \Big|_0^1 = 1 - 2/3 = 1/3]$$

CONTEST 28 (Q/F) ROUND 1B

Find the quadratic inequality with the given solution set

1 $\{x: -7 < x < 3\}$

ANSWER: $(x + 7)(x - 3) < 0$, or $x^2 + 4x - 21 < 0$

$$[(x + 7)(x - 3) < 0, x^2 + 7x - 3x - 21 = x^2 + 4x - 21 < 0]$$

2 $\{x: x > 5, \text{ or } x < 3\}$

ANSWER: $(x - 5)(x - 3) > 0$, or $x^2 - 8x + 15 > 0$

$$[(x - 5)(x - 3) > 0, x^2 - 5x - 3x + 15 = x^2 - 8x + 15 > 0]$$

3 $\{x: -6 < x < 6\}$

ANSWER: $(x - 6)(x + 6) < 0$, or $x^2 - 36 < 0$

$$[(x + 6)(x - 6) < 0, x^2 + 6x - 6x - 36 = x^2 - 36 < 0]$$

CONTEST 28 (Q/F) ROUND 1C

Find in radians

1 $\sec^{-1} 2$

ANSWER: $\pi/3$

$[\sec^{-1} 2 = \cos^{-1} (1/2) = \pi/3]$

2 $\operatorname{cosec}^{-1}(-1)$

ANSWER: $-\pi/2$

$[\operatorname{cosec}^{-1}(-1) = \sin^{-1}(-1) = -\pi/2]$

3 $\cot^{-1} (1/\sqrt{3})$

ANSWER: $\pi/3$

$[\cot^{-1} (1/\sqrt{3}) = \tan^{-1} \sqrt{3} = \pi/3]$

CONTEST 28 ROUND 2: SPEED RACE

1 Find the coordinates of the vertex of the quadratic function

$y = -3x^2 + 18x - 23$

ANSWER: (3, 4)

$[-3(x^2 - 6x) - 23 = -3(x^2 - 6x + 9) + 27 - 23 = -3(x - 3)^2 + 4, \text{ hence } (3, 4)]$

2 Find the domain of the function $f(x) = x^2 - 3x + 2$

ANSWER: {x: x is a real number} or {real numbers}

3 Solve for x from the equation $\log(x^2 - 9x) = 1$

ANSWER: x = 10, -1

$[x^2 - 9x = 10, x^2 - 9x - 10 = (x - 10)(x + 1) = 0, x = 10, -1]$

CONTEST 28 ROUND 5: RIDDLE

1 I am simply a line segment.

2 I am associated with polygons and polyhedrons.

3 I am not an edge.

4 I lie completely inside the polygon or the polyhedron

5 In a polygon, I connect non-adjacent vertices.

6 In a polyhedron, I connect vertices not in the same face.
Who am I?

ANSWER: DIAGONAL

CONTEST 28 ROUND 4A: TRUE OR FALSE

1 $\cos(\pi + x) = \cos(\pi - x)$

ANSWER: TRUE

2 $\sin(\pi + x) = \sin(\pi - x)$

ANSWER: FALSE

3 $\tan(\pi + x) = \tan(\pi - x)$

ANSWER: FALSE

CONTEST 28 ROUND 4B: TRUE OR FALSE

1 $\sqrt{5} + \sqrt{5} < \sqrt{10}$

ANSWER: FALSE

2 $\sqrt{3} + \sqrt{3} > \sqrt{6}$

ANSWER: TRUE

3 For any positive number a, $2\sqrt{a} > \sqrt{(2a)}$

ANSWER: TRUE

CONTEST 29 ROUND 1A

Find the equation of the perpendicular bisector of the line segment AB given

1 A(1, 2) and B(3, 1)

ANSWER: $y = 2x - 5/2$

$$[(x - 1)^2 + (y - 2)^2 = (x - 3)^2 + (y - 1)^2, \\ -2x + 1 - 4y + 4 = -6x + 9 + -2y + 1, 4x + 5 = 2y + 10, y = 2x - 5/2]$$

2 A(-2, -1) and B(1, 2)

ANSWER: $y = -x$

$$[(x + 2)^2 + (y + 1)^2 = (x - 1)^2 + (y - 2)^2, \\ 4x + 4 + 2y + 1 = -2x + 1 - 4y + 4, 6x + 5 = -6y + 5, y = -x]$$

3 A(2, -1) and B(-2, 2)

ANSWER: $y = (4/3)x + 1/2$

$$[(x - 2)^2 + (y + 1)^2 = (x + 2)^2 + (y - 2)^2, \\ -4x + 4 + 2y + 1 = 4x + 4 - 4y + 4, 6y + 5 = 8x + 8, y = (4/3)x + 1/2]$$

CONTEST 29 ROUND 1B

Given that b is positive and $a > 4$, find

$$1 \cos(\tan^{-1}(b/3))$$

ANSWER: $3/\sqrt{b^2 + 9}$

[Let $\theta = \tan^{-1}(b/3)$, $\tan \theta = b/3$, $\cos \theta = 3/\sqrt{b^2 + 9}$]

$$2 \sin(\cos^{-1}(4/a))$$

ANSWER: $\sqrt{a^2 - 16}/a$

[Let $\theta = \cos^{-1}(4/a)$, $\cos \theta = 4/a$, $\sin \theta = \sqrt{a^2 - 16}/a$]

$$3 \sin(\cot^{-1}(a/b))$$

ANSWER: $b/\sqrt{a^2 + b^2}$

[Let $\theta = \cot^{-1}(a/b)$, $\cot \theta = a/b$, $\sin \theta = b/\sqrt{a^2 + b^2}$]

CONTEST 29 ROUND 1C

Express a in terms of b given

$$1 \ a = \log_8 27 \text{ and } b = \log_4 9 \quad \textbf{ANSWER: } a = b$$

[$a = \log_2 27 / \log_2 8 = (\log_2 3^3) / 3 = \log_2 3$, $b = \log_2 3^2 / \log_2 2^2 = \log_2 3$,
hence $a = b$]

$$2 \ a = \log_4 625, b = \log_8 125 \quad \textbf{ANSWER: } a = 2b$$

[$a = 4(\log_2 5) / \log_2 4 = 2 \log_2 5$, $b = (3 \log_2 5) / 3 = \log_2 5$ hence $a = 2b$]

$$3 \ a = \log_3 64, b = \log_9 256 \quad \textbf{ANSWER: } a = 3b/2$$

[$a = (\log_3 4^3) = 3 \log_3 4$, $b = (\log_3 4^4) / \log_3 9 = 2 \log_3 4$, $a/b = 3/2$, $a = 3b/2$]

CONTEST 29 ROUND 2: SPEED RACE

1 Find the area of a rhombus if one diagonal is of length 16 cm and a side has length 17 cm.

ANSWER: 240 cm²

[Let other diagonal have length $2x$, $x^2 + 8^2 = 17^2$, $x^2 = 289 - 64 = 225$, $x = 15$
Area = $\frac{1}{2} d_1 d_2 = \frac{1}{2} (16 \times 30) = 240 \text{ cm}^2$]

2 A sequence is defined by the recurrence relation $U_{n+1} = 2U_n - 3n$, $U_1 = 1$.
Find the next two terms.

ANSWER: $U_2 = -1$, $U_3 = -8$

[$U_2 = 2U_1 - 3(1) = 2 - 3(1) = -1$, $U_3 = 2U_2 - 3(2) = -2 - 6 = -8$]

3 Simplify the trigonometric expression $(\cos x + \sin x)^2 - \sin 2x$

ANSWER: 1

[$\sin^2 x + \cos^2 x + 2\sin x \cos x - \sin 2x = 1 + \sin 2x - \sin 2x = 1$]

CONTEST 29 ROUND 5: RIDDLE

1 I am a quadrilateral.

2 You will find me inside a circle.

3 I have a circumscribing circle.

4 My vertices lie on a circle.

5 My opposite angles are supplementary.

6 A rectangle is an example of me.

Who am I?

ANSWER: CYCLIC QUADRILATERAL

CONTEST 29 ROUND 4A: TRUE OR FALSE

The solution set of the inequality $(x + 3)(x - 2) > 0$ is

1 $\{x: -2 < x < 3\}$

ANSWER: FALSE

2 $\{x: x < -2, \text{ or } x > 3\}$

ANSWER: FALSE

3 $\{x: -3 < x < 2\}$

ANSWER: FALSE

CONTEST 29 ROUND 4B: TRUE OR FALSE

1 $x^3 + 2x^2 - 4x - 8 = (x + 2)(x - 2)^2$

ANSWER: FALSE

2 $x^3 + 2x^2 - 4x - 8 = (x - 2)(x^2 + 4)$

ANSWER: FALSE

3 $x^3 + 2x^2 - 4x - 8 = (x - 2)(x^2 - 4)$

ANSWER: FALSE

CONTEST 30 ROUND 1A

Solve the equation for real values of x

1 $|x^2 - 2x + 1| = 4$

ANSWER: $x = 3, -1$

$[(x - 1)^2 = 4, x - 1 = \pm 2, x = 1 \pm 2, x = 3, -1]$

2 $|x^2 - 3x + 5| = 9$

ANSWER: $x = 4, -1$

$[x^2 - 3x + 5 = \pm 9, x^2 - 3x + 5 = 9, x^2 - 3x - 4 = (x - 4)(x + 1) = 0,$
 $x = 4, -1, x^2 - 3x + 5 = -9, x^2 - 3x + 14 = 0$ has no real roots, as $b^2 - 4ac < 0]$

3 $|x^2 + 2x + 5| = 5$

ANSWER: $x = 0, -2$

$[x^2 + 2x + 5 = \pm 5, x^2 + 2x + 5 = 5, x^2 + 2x = 0, x = 0, -2; x^2 + 2x + 5 = -5$
 $x^2 + 2x + 10 = 0$ has no real roots, as $b^2 - 4ac < 0]$

CONTEST 30 ROUND 1B

Find a function $g(x)$ such that $f(g(x)) = x$, given

1 $f(x) = 3/(2x + 1)$

ANSWER: $g(x) = (3 - x)/2x$

$[g(x) = f^{-1}(x), \text{ let } y = 3/(2x + 1), x = 3/(2y + 1), x(2y + 1) = 3, 2xy = 3 - x,$
 $y = (3 - x)/2x, g(x) = f^{-1}(x) = (3 - x)/2x]$

2 $f(x) = (x + 1)/(2x + 1)$

ANSWER: $g(x) = (1 - x)/(2x - 1)$

$[g(x) = f^{-1}(x), \text{ let } y = (x + 1)/(2x + 1), x = (y + 1)/(2y + 1),$
 $x(2y + 1) = y + 1, y(2x - 1) = 1 - x, g(x) = f^{-1}(x) = (1 - x)/(2x - 1)]$

3 $f(x) = (2x + 3)/(x + 2)$

ANSWER: $g(x) = (3 - 2x)/(x - 2)$

$[\text{Let } y = (2x + 3)/(x + 2), x = (2y + 3)/(y + 2), x(y + 2) = (2y + 3),$
 $y(x - 2) = (3 - 2x), y = (3 - 2x)/(x - 2), g(x) = f^{-1}(x) = (3 - 2x)/(x - 2)]$

CONTEST 30 ROUND 1C

Find the coordinates of the point(s) of inflexion of the given curve.

1 $y = x^4 - 6x^3 + 12x^2$ **ANSWER: (1, 7) and (2, 16)**
[$dy/dx = 4x^3 - 18x^2 + 24x$, $d^2y/dx^2 = 12x^2 - 36x + 24 = 12(x^2 - 3x + 2) = 0$,
 $x = 1, 2$. For $x = 1$, $y = 1 - 6 + 12 = 7$, and for $x = 2$, $y = 16 - 48 + 48 = 16$,
hence (1, 7) and (2, 16)]

2 $y = x^4 + 4x^3 + 16$ **ANSWER: (0, 16) and (-2, 0)**
[$dy/dx = 4x^3 + 12x^2$, $d^2y/dx^2 = 12x^2 + 24x = 12x(x + 2) = 0$, $x = 0, -2$.
For $x = 0$, $y = 16$ and for $x = -2$, $y = 16 - 32 + 16 = 0$, hence (0, 16), (-2, 16)]

3 $y = x^4 - 6x^2 + 2x$ **ANSWER: (1, -3) and (-1, -7)**
[$dy/dx = 4x^3 - 12x + 2$, $d^2y/dx^2 = 12x^2 - 12 = 12(x^2 - 1) = 0$, $x = \pm 1$. For $x = 1$,
 $y = 1 - 6 + 2 = -3$ and for $x = -1$, $y = 1 - 6 - 2 = -7$, hence (1, -3) and (-1, -7)]

CONTEST 30 ROUND 2: SPEED RACE

1 Find the first three terms of the sequence defined by $U_n = \cos(n\pi/6)$
ANSWER: $U_1 = \sqrt{3}/2$, $U_2 = 1/2$, $U_3 = 0$
[$U_1 = \cos(\pi/6) = \sqrt{3}/2$, $U_2 = \cos(\pi/3) = 1/2$, $U_3 = \cos(3\pi/6) = \cos(\pi/2) = 0$]

2 Find the solution set of the compound inequality $-15 < 3x - 6 < 9$
ANSWER: $\{x: -3 < x < 5\}$
[$-15 < 3x - 6 < 9$, $-9 < 3x < 15$, $-3 < x < 5$]

3 Given the matrices $A = \begin{pmatrix} 2 & -3 \\ 4 & 6 \end{pmatrix}$ and $B = \begin{pmatrix} 3 & 5 \\ 4 & -2 \end{pmatrix}$ find the matrix AB.

ANSWER: $AB = \begin{pmatrix} -6 & 16 \\ 36 & 8 \end{pmatrix}$ [$AB = \begin{pmatrix} 2 & -3 \\ 4 & 6 \end{pmatrix} \begin{pmatrix} 3 & 5 \\ 4 & -2 \end{pmatrix} = \begin{pmatrix} 6-12 & 10+6 \\ 12+24 & 20-12 \end{pmatrix} = \begin{pmatrix} -6 & 16 \\ 36 & 8 \end{pmatrix}$]

CONTEST 30 ROUND 5: RIDDLE

- 1 We are a set of points.
- 2 We are associated with a unique straight line.
- 3 Any two of us define the straight line.
- 4 Any three of us lie on the same straight line.
- 5 Generally we are points that lie on the same straight line.
- 6 If points on the same plane are coplanar points, then who are we?

ANSWER: COLLINEAR POINTS (Accept COLLINEAR)

CONTEST 30 ROUND 4A: TRUE OR FALSE

- 1 A function $f(x)$ is an even function if its graph is symmetric about the origin

ANSWER: FALSE

- 2 A function is odd if its graph is symmetric about the origin.

ANSWER: TRUE

- 3 A function is even if its graph is symmetric about the x-axis

ANSWER: FALSE

CONTEST 30 ROUND 4B: TRUE OR FALSE

1 A matrix has a determinant only if it is a square matrix.

ANSWER: TRUE

2 Every square matrix has an inverse.

ANSWER: FALSE

3 A matrix has an inverse if its determinant is zero.

ANSWER: FALSE

CONTEST 31 ROUND 1A

Express p^2 in terms of q given that the equation $x^2 - px + q = 0$ has

1 roots k , and $(k + 1)$

ANSWER: $p^2 = 4q + 1$ [$p = 2k + 1$, $q = k(k + 1)$, $k = (p - 1)/2$,
 $q = ((p - 1)/2)((p + 1)/2) = (p^2 - 1)/4$, $p^2 = 4q + 1$]

2 has roots $(k - 1)$ and $(k + 2)$

ANSWER: $p^2 = 4q + 9$ [$2k + 1 = p$, $k = (p - 1)/2$, $q = (k - 1)(k + 2) =$
 $(p - 3)(p + 3)/4$, $p^2 - 9 = 4q$, $p^2 = 4q + 9$]

3 has roots $(k - 2)$ and $(k + 2)$

ANSWER: $p^2 = 4q + 16$ [$2k = p$, $k = p/2$, $q = (k - 2)(k + 2) = k^2 - 4 =$
 $(p/2)^2 - 4$, $q = p^2/4 - 4$, $p^2 = 4q + 16$]

CONTEST 31 ROUND 1B

Find a relation between the scalars x and y given that

1 the vectors $\mathbf{a} = (x - 2)\mathbf{i} - 3\mathbf{j}$ and $\mathbf{b} = 5\mathbf{i} + y\mathbf{j}$ are perpendicular,

ANSWER: $5x - 3y = 10$, or $y = 5x/3 - 10/3$

[$\mathbf{a} \cdot \mathbf{b} = 5(x - 2) - 3y = 0$, $5x - 3y = 10$, or $y = 5x/3 - 10/3$]

2 the vectors $\mathbf{a} = 2\mathbf{i} + y\mathbf{j}$ and $\mathbf{b} = x\mathbf{i} + 5\mathbf{j}$ are parallel,

ANSWER: $xy = 10$, or $y = 10/x$

[$2/y = x/5$, $xy = 10$, or $y = 10/x$]

3 the vectors $\mathbf{a} = 4x\mathbf{i} + y\mathbf{j}$ and $\mathbf{b} = x\mathbf{i} - y\mathbf{j}$ are perpendicular.

ANSWER: $4x^2 - y^2 = 0$ or $y^2 = 4x^2$, or $y = \pm 2x$

[$\mathbf{a} \cdot \mathbf{b} = 4x^2 - y^2 = 0$, $y^2 = 4x^2$]

CONTEST 31 ROUND 1C

Find the coordinates of the point(s) on the given curve at which the gradient m has the given value

1 $y = 1 + 1/x$, and $m = -1$

ANSWER: (1, 2) and (-1, 0) [$dy/dx = -1/x^2 = -1$, $x^2 = 1$, $x = \pm 1$,

$x = 1$ and $y = 2$, $x = -1$ and $y = 0$, (1, 2) and (-1, 0)]

2 $y = 4x + 1/x$, and $m = 0$

ANSWER: (1/2, 4) and (-1/2, -4) [$dy/dx = 4 - 1/x^2 = 0$, $x^2 = 1/4$, $x = \pm 1/2$, for $x =$

$1/2$, $y = 4$, for $x = -1/2$, $y = -4$, hence (1/2, 4) and (-1/2, -4)]

3 $y = x^2 + 1/x^2$, and $m = 0$

ANSWER: (1, 2) and (-1, 2)

[$dy/dx = 2x - 2/x^3 = 0$, $x^4 = 1$, $x = \pm 1$, for $x = 1$, $y = 2$, for $x = -1$, $y = 2$]

CONTEST 31 ROUND 2: SPEED RACE

1 Find the degree measures of the angles of a triangle if they are in the ratio

4 : 3 : 2. **ANSWER: 80°, 60°, 40°**

[$4x + 3x + 2x = 9x = 180$, $x = 20$, $4x = 80$, $3x = 60$ and $2x = 40$]

2 The velocities of particles A and B are $v_A = (xi + 7j)$ m/s and

$v_B = (5i + yj)$ m/s respectively. Evaluate x and y if the velocity of A relative to B is

$(3i + 4j)$ m/s **ANSWER: $x = 8, y = 3$** [$(xi + 7j) - (5 + yj) = (x - 5)i + (7 - y)j$
 $= 3i + 4j$, $(x - 5) = 3$, $x = 8$, $(7 - y) = 4$, $y = 3$]

3 Find the zeros of the function $f(x) = x^3 + 3x^2 - x - 3$

ANSWER: $x = \pm 1, -3$

[$x^2(x + 3) - 1(x + 3) = (x + 3)(x^2 - 1) = 0$, $x = \pm 1, -3$]

CONTEST 31 ROUND 5: RIDDLE

.

1 We are a pair of real numbers.

2 We may be roots of a quadratic equation.

3 Our sum is -8.

4 Our product is -48.

5 We have opposite signs.

Who are we?

ANSWER: -12, 4

CONTEST 31 ROUND 4A: TRUE OR FALSE

1 If opposite angles of a square are complementary, then it is a triangle.

ANSWER: TRUE

2 If the sum of the angles of a quadrilateral is 400° , then it is a rhombus.

ANSWER: TRUE

3 If the sum of the angles of a triangle is a straight angle, then it is a square.

ANSWER: FALSE

CONTEST 31 ROUND 4B: TRUE OR FALSE

1 If the sum of two real numbers is an irrational number, then at least one of the numbers is an irrational number.

ANSWER: TRUE

2 If the product of two real numbers is a rational number, then both numbers are rational numbers.

ANSWER: FALSE

3 The product of 2 irrational numbers is an irrational number.

ANSWER: FALSE

CONTEST 32 ROUND 1A

A particle moves in a straight line such that at time t seconds, its displacement from a fixed point O on the line is given by $x = (t^3 - 2t^2 + 5t)$ meters. Find

1 its velocity v when $t = 2$

ANSWER: $v = 9 \text{ m/s}$

[$v = dx/dt = 3t^2 - 4t + 5$, for $t = 2$, $v = 12 - 8 + 5 = 9 \text{ m/s}$]

2 its acceleration a when $t = 2$

ANSWER: $a = 8 \text{ ms}^{-2}$

[$a = dv/dt = 6t - 4$, for $t = 2$, $a = 12 - 4 = 8 \text{ m/s}^2$]

3 the time when the particle is at the point O

ANSWER: $t = 0$

[$t(t^2 - 2t + 5) = 0$, $t = 0$, $t^2 - 2t + 5 = 0$ has no real solution]

CONTEST 32 ROUND 1B

Two fair dice are thrown. Find the probability that the total score is

1. 4 or 8

ANSWER: $2/9$

[$A = \{(1, 3), (3, 1), (2, 2), (2, 6), (6, 2), (3, 5), (5, 3), (4, 4)\}$, $P(A) = 8/36 = 2/9$]

2. 5 or 10

ANSWER: $5/36$

[$B = \{(2, 3), (3, 2), (4, 6), (6, 4), (5, 5)\}$ $P(B) = 5/36$]

3. 7 or 9

ANSWER: $5/18$

[$C = \{(1, 6), (6, 1), (2, 5), (5, 2), (3, 4), (4, 3), (3, 6), (6, 3), (4, 5), (5, 4)\}$
 $P(C) = 10/36 = 5/18$]

CONTEST 32 ROUND 1C

Factorise completely

1 $(x - 1)^4 - (x - 1)^2$

ANSWER: $x(x - 1)^2(x - 2)$

$[(x - 1)^2 ((x - 1)^2 - 1)] = (x - 1)^2(x - 1 - 1)(x - 1 + 1) = x(x - 1)^2(x - 2)]$

2 $(x + 1)^4 + (x + 1)^2$

ANSWER: $(x + 1)^2(x^2 + 2x + 2)$ accept $(x + 1)^2((x + 1)^2 + 1)$

$[(x + 1)^2 ((x + 1)^2 + 1) = (x + 1)^2 (x^2 + 2x + 2)]$

3 $(x + a)^3 + x^3$

ANSWER: $(2x + a)(x^2 + ax + a^2)$

$[(x + a + x)((x + a)^2 - x(x + a) + x^2) = (2x + a)((x + a)^2 - ax)$
 $= (2x + a)(x^2 + ax + a^2)]$

CONTEST 32 ROUND 2: SPEED RACE

1 Find the coordinates of the point $P(x, y)$ whose image under the linear transformation $T: (x, y) \rightarrow (3x - y, x + y)$ is the point $Q(0, -4)$

ANSWER: $(-1, -3)$

$[3x - y = 0, x + y = -4, y = 3x, 4x = -4, x = -1, y = 3x = -3]$

2 Solve for x from the equation $(\log x)^2 - \log x = 2$

ANSWER: $x = 100, 1/10$

$[(\log x - 2)(\log x + 1) = 0, \log x = 2, x = 10^2 = 100, \log x = -1, x = 10^{-1} = 1/10]$

3 Solve the equation $\sin 2x - 2\cos x = 0$ for $0 \leq x \leq \pi$

ANSWER: $x = \pi/2$

$[2\sin x \cos x - 2\cos x = 2\cos x(\sin x - 1) = 0, \cos x = 0, \text{ then } x = \pi/2, \sin x = 1 \text{ then } x = \pi/2]$

CONTEST 32 ROUND 5: RIDDLE

1 I am a geometrical measure.

2 I am associated with angles.

3 I relate to the least amount of rotation necessary from the overlapping position to the final angle.

4 The instrument used is a protractor.

5 My units are degrees or radians.
Who am I?

ANSWER: MEASURE OF AN ANGLE

CONTEST 32 ROUND 4A: TRUE OR FALSE

In the following, k is a constant and x , y , z , t are variables

1 If $xy = k$, then the variation is a joint variation.

ANSWER: FALSE

2 If $y = kxz$, the variation is a joint variation.

ANSWER: TRUE

3 If $y = kxz/t^2$, the variation is a combined variation.

ANSWER: TRUE

CONTEST 32 ROUND 4B: TRUE OR FALSE

$$1 \log [(x^3 + y^3)/(x + y)] = \log(x^2 - xy + y^2)$$

ANSWER: TRUE

$$2 \log[(x^3 - y^3)/(x^2 + xy + y^2)] = \log(x - y)$$

ANSWER: TRUE

$$3 \log[(x^4 - y^4)/(x^2 + y^2)] = \log(x^2 - y^2)$$

ANSWER: TRUE

CONTEST 33 ROUND 1A

Evaluate the trigonometric expression and rationalize the denominator

$$1 (\sec 60^\circ - \operatorname{cosec} 60^\circ) / \sec 45^\circ$$

$$\textbf{ANSWER : } (\sqrt{3} - 1)\sqrt{6}/3 \text{ or } (3\sqrt{2} - \sqrt{6})/3 \text{ or } \sqrt{2} - \sqrt{6}/3$$

$$[(2 - 2/\sqrt{3})/\sqrt{2} = 2(\sqrt{3} - 1)/\sqrt{6} = (\sqrt{3} - 1)\sqrt{6}/3 \text{ or } (3\sqrt{2} - \sqrt{6})/3, (\sqrt{2} - \sqrt{6}/3)]$$

$$2 (\operatorname{cosec} 30^\circ - \sec 45^\circ) / \operatorname{cosec} 60^\circ$$

$$\textbf{ANSWER: } (2 - \sqrt{2})\sqrt{3}/2 \text{ or } (2\sqrt{3} - \sqrt{6})/2 \text{ or } (\sqrt{3} - \sqrt{6}/2)$$

$$[(2 - \sqrt{2})/(2/\sqrt{3}) = (2 - \sqrt{2})\sqrt{3}/2 \text{ or } (2\sqrt{3} - \sqrt{6})/2, (\sqrt{3} - \sqrt{6}/2)]$$

$$3 (\sec 30^\circ - \sec 45^\circ) / \operatorname{cosec} 30^\circ$$

$$\textbf{ANSWER: } (2\sqrt{3} - 3\sqrt{2})/6, \text{ or } (\sqrt{3}/3 - \sqrt{2}/2)$$

$$[(2/\sqrt{3} - \sqrt{2})/(2) = (2 - \sqrt{6})/2\sqrt{3} = (2\sqrt{3} - 3\sqrt{2})/6, (\sqrt{3}/3 - \sqrt{2}/2)]$$

CONTEST 33 ROUND 1B

20 different English books and 15 different French books are to be placed on a shelf.
Find the number of ways this can be done if
(Leave answer in factorial notation)

1 the English books are to be together,

ANSWER: $20! \times 16!$

[arrange 20 English books in $20!$ ways, combine with 15 French books in $16!$ ways,
hence $20! \times 16!$]

2 the French books are to be together,

ANSWER: $15! \times 21!$

[arrange 15 French books in $15!$ ways, combine with 20 English books in $21!$ ways,
hence $15! \times 21!$]

3 the English books are to be together and the French books are also to be together.

ANSWER: $20! \times 15! \times 2!$

[arrange English books in $20!$ ways, arrange French books in $15!$ ways and arrange the
two blocks of English books and French books in $2!$ ways, hence $20! \times 15! \times 2!$]

CONTEST 33 ROUND 1C

Find the coordinates of the stationary points of the given curve

1 $y = x^3 - 12x + 20$ **ANSWER: $(2, 4), (-2, 36)$**

[$dy/dx = 3x^2 - 12 = 0$, $x^2 = 4$, $x = \pm 2$, for $x = 2$, $y = 8 - 24 + 20 = 4$,
hence $(2, 4)$, for $x = -2$, $y = -8 + 24 + 20 = 36$, hence $(-2, 36)$]

2 $y = 2x^3 + 3x^2 - 12x$ **ANSWER: $(1, -7), (-2, 20)$**

[$dy/dx = 6x^2 + 6x - 12 = 6(x^2 + x - 2) = 6(x + 2)(x - 1) = 0$, $x = 1, -2$, for $x = 1$
 $y = 2 + 3 - 12 = -7$, $(1, -7)$; for $x = -2$, $y = -16 + 12 + 24 = 20$ $(-2, 20)$]

3 $y = x^3 - 3x^2 + 6$ **ANSWER: $(0, 6), (2, 2)$**

[$dy/dx = 3x^2 - 6x = 3x(x - 2) = 0$, $x = 0, 2$, for $x = 0$, $y = 6$, $(0, 6)$; for
 $x = 2$, $y = 8 - 12 + 6 = 2$ hence $(2, 2)$]

CONTEST 33 ROUND 2: SPEED RACE

1 Evaluate the definite integral $\int_0^1 (x^2 - 4)^2 dx$

ANSWER: 203/15
$$\left[\int_0^1 (x^4 - 8x^2 + 16) dx = x^5 / 5 - 8x^3 / 3 + 16x \right]_0^1$$

 $= 1/5 - 8/3 + 16 = (3 - 40 + 240)/15 = 203/15]$

2 Find the values of b and c if the inequality $x^2 + bx + c > 0$ has solution set $\{x: x > 1, \text{ or } x < -2\}$ **ANSWER: b = 1, c = -2**
 $[(x - 1)(x + 2) = (x^2 + x - 2) > 0, \text{ hence , } b = 1, c = -2]$

3 Obtain an equation in x and y by eliminating θ from the pair of equations
 $x = 3\sin\theta, y = 4\cos\theta$
ANSWER: $x^2/9 + y^2/16 = 1$
 $[\sin\theta = x/3, \cos\theta = y/4, \cos^2\theta + \sin^2\theta = 1, x^2/9 + y^2/16 = 1]$

CONTEST 33 ROUND 4A: TRUE OR FALSE

Two matrices A and B may be added if

1 they have the same number of columns,
ANSWER: FALSE

2 they have the same number of rows,
ANSWER: FALSE

3 the matrices are both square matrices.
ANSWER: FALSE

CONTEST 33 ROUND 4B: TRUE OR FALSE

1 For a quadratic equation, if the discriminant is positive, the roots are rational

ANSWER: FALSE

2 If the roots of a quadratic equation are rational, the discriminant is non-negative.

ANSWER: TRUE

3 If the discriminant is negative, a quadratic equation has no roots.

ANSWER: FALSE

CONTEST 33 ROUND 5: RIDDLE

1 I am a function.

2 I am a polynomial function.

3 My graph intersects the x-axis at least once and at most three times.

4 I have at most two stationary points, a maximum and a minimum.

5 I always have one point of inflexion.

6 I have at most three zeros.

Who am I?

ANSWER: CUBIC FUNCTION

CONTEST 34 ROUND 1A

Evaluate the given limit by rationalizing the numerator.

1 $\lim_{x \rightarrow 0} \frac{1 - \sqrt{1-x}}{x}$

ANSWER: 1/2

$[(1 - 1 + x)/x(1 + \sqrt{1-x}) = 1/(1 + \sqrt{1-x})]$, hence limit = $1/(1 + 1) = \frac{1}{2}$]

2 $\lim_{x \rightarrow 2} \frac{(\sqrt{x+2}-2)}{x^2-2x}$

ANSWER: 1/8 $[(x + 2 - 4)/x(x - 2)(\sqrt{x+2} + 2) = 1/x(\sqrt{x+2} + 2)]$, hence limit is $1/2(2 + 2) = 1/8$

3 $\lim_{x \rightarrow 0} \frac{(\sqrt{1+2x}-1)}{x}$

ANSWER: 1 $[(1 + 2x - 1)/x(\sqrt{1+2x} + 1) = 2/(\sqrt{1+2x} + 1)]$, hence limit is $2/(1 + 1) = 1$

CONTEST 34 ROUND 1B

Solve the equation for x in the interval $0 \leq x \leq \pi/2$. Give answer in radians.

1 $\log (2\sin x) = 0$

ANSWER: $x = \pi/6$ $[2\sin x = 1, \sin x = 1/2, x = \pi/6]$

2 $\log (2\cos x) = 0$

ANSWER: $x = \pi/3$ $[2\cos x = 1, \cos x = 1/2, x = \pi/3]$

3 $\log (\tan x) = 0$

ANSWER: $x = \pi/4$ $[\tan x = 1, x = \pi/4]$

CONTEST 34 ROUND 1C

A particle is travelling in a straight line with a uniform acceleration. Find

1 the acceleration if it travels a distance of 60 m while velocity changes from 5 ms^{-1} to 25 ms^{-1}

ANSWER: 5 ms^{-2}

$$[a = (v^2 - u^2)/2s = (625 - 25)/2(60) = 600/120 = 5 \text{ ms}^{-2}]$$

2 the final velocity if it travels a distance of 100 m from an initial velocity of 10 ms^{-1} with a uniform acceleration of 2 ms^{-2} .

ANSWER: $10\sqrt{5} \text{ ms}^{-1}$

$$[v^2 = u^2 + 2as = 100 + 2(100)2 = 500, v = 10\sqrt{5} \text{ ms}^{-1}]$$

3 the initial velocity if it travels a distance of 50 m while reaching a velocity of 15 ms^{-1} with uniform acceleration of 1 ms^{-2} .

ANSWER: $5\sqrt{5} \text{ ms}^{-1}$

$$[u^2 = v^2 - 2as = 225 - 2(50)1 = 125, v = 5\sqrt{5} \text{ ms}^{-1}]$$

CONTEST 34 ROUND 2: SPEED RACE

1 Find the solution set of the rational inequality $2/(x - 2) < 0$.

ANSWER: $\{x: x < 2\}$

$$[(x - 2)/2 < 0, x - 2 < 0, x < 2]$$

2 Find the sum of the interior angles of a polygon of 13 sides in radians.

ANSWER: 11π radians

$$[(13 - 2)\pi = 11\pi \text{ radians}]$$

3 Find the volume of the solid of rotation when the portion of the curve $y = (3x + 2)$ between $x = 0$ and $x = 1$ is rotated completely about the x-axis.

ANSWER: 13π cubic units

$$\left[\pi \int_0^1 (3x + 2)^2 dx = \pi \int_0^1 (9x^2 + 12x + 4) dx = \pi (3x^3 + 6x^2 + 4x) \Big|_0^1 = 13\pi \right]$$

CONTEST 34 ROUND 5: RIDDLE

1 I am a vector.

2 I have a fixed length.

3 However I have no fixed direction.

4 Every vector is a scalar multiple of me.

5 Examples of me are the vectors **i** and **j**.

6 I am a vector of unit length.
Who am I?

ANSWER: UNIT VECTOR

CONTEST 34 ROUND 4A: TRUE OR FALSE

1 $\sqrt{(-2)^2} = -2$

ANSWER: FALSE

2 $\sqrt[3]{-8} = -2$

ANSWER: TRUE

3 $\sqrt[4]{-16} = -2$

ANSWER: FALSE

CONTEST 34 ROUND 4B: TRUE OR FALSE

1 $(x^3 - 1)/(x^2 - 1) = (x^2 + x + 1)/(x + 1)$

ANSWER: TRUE

2 $(x^3 + 1)/(x + 1) = x^2 - x + 1$

ANSWER: TRUE

3 $(x^3 + 1)/(x^2 + 1) = (x^2 - x + 1)/(x - 1)$

ANSWER: FALSE

CONTEST 35 ROUND 1A

Find in radians

1 $\cos^{-1}(-1/2)$

ANSWER: $2\pi/3$

$[\cos^{-1}(-1/2) = \pi - \cos^{-1}(1/2) = \pi - \pi/3 = 2\pi/3]$

2 $\tan^{-1}(-1/\sqrt{3})$

ANSWER: $-\pi/6$

$[\tan^{-1}(-1/\sqrt{3}) = -\tan^{-1}(1/\sqrt{3}) = -\pi/6]$

3 $\sin^{-1}(-\sqrt{2}/2)$

ANSWER: $-\pi/4$

$[\sin^{-1}(-\sqrt{2}/2) = -\sin^{-1}(\sqrt{2}/2) = -\pi/4]$

CONTEST 35 ROUND 1B

α and β are the roots of the quadratic equation $2x^2 - 5x - 4 = 0$. Evaluate

1 $(\alpha + 1)(\beta + 1)$

ANSWER: $3/2$

$[\alpha\beta + (\alpha + \beta) + 1 = -4/2 + 5/2 + 1 = -1 + 5/2 = 3/2]$

2 $\alpha^3 + \beta^3$

ANSWER: $245/8$ $[(\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) = (5/2)^3 - 3(-2)(5/2) = 125/8 + 15 = 245/8]$

3 $(\alpha - \beta)^2$

ANSWER: $57/4$

$[(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta = (5/2)^2 - 4(-2) = 25/4 + 8 = 57/4]$

CONTEST 35 ROUND 1C

Find the equation of a function if its gradient is given by

1 $dy/dx = 6x^2 - 8x + 5$ and it passes through the point A(1, 4),

ANSWER: $y = 2x^3 - 4x^2 + 5x + 1$ $[y = 2x^3 - 4x^2 + 5x + C;$

for $x = 1, 2 - 4 + 5 + C = 4, C = 1, y = 2x^3 - 4x^2 + 5x + 1]$

2 $dy/dx = 8x^3 + 3x^2 - 10x$ and it passes through the point B(-1, 0),

ANSWER: $y = 2x^4 + x^3 - 5x^2 + 4$ $[y = 2x^4 + x^3 - 5x^2 + C;$

for $x = -1, 2 - 1 - 5 + C = 0, C = 4, \text{ and } y = 2x^4 + x^3 - 5x^2 + 4]$

3 $dy/dx = 5x^4 - 6x^2 + 2x$ and it passes through the point C(1, 2)

ANSWER: $y = x^5 - 2x^3 + x^2 + 2$ $[y = x^5 - 2x^3 + x^2 + C,$

for $x = 1, 2 = 1 - 2 + 1 + C, C = 2, y = x^5 - 2x^3 + x^2 + 2]$

CONTEST 35 ROUND 2: SPEED RACE

1 Find the value of x if the matrix $A = \begin{pmatrix} x+2 & -4 \\ -2 & x-2 \end{pmatrix}$ has no inverse.

ANSWER: $x = \pm 2\sqrt{3}$

[$\det A = (x+2)(x-2) - 8 = x^2 - 4 - 8 = 0$, $x^2 = 12$, $x = \pm 2\sqrt{3}$]

2 A binary operation $*$ is defined on the set R of real numbers by $a*b = a + b - ab$ (Assume $a, b \neq 1$). Find the identity e .

ANSWER: $e = 0$ [$e*a = e + a - ea = a$, $e(1-a) = 0$, $e = 0$, since $a \neq 1$]

3 Find the coordinates of the vertex as well as the equation of the axis of symmetry of the quadratic curve $y = 3x^2 - 12x + 10$

ANSWER: $(2, -2)$ and $x = 2$ [$3(x^2 - 4x) + 10 = 3(x^2 - 4x + 4) + 10 - 12 = 3(x-2)^2 - 2$, hence vertex is $(2, -2)$ and axis of symmetry is $x = 2$]

CONTEST 35 ROUND 4A: TRUE OR FALSE

1 The centroid of a triangle is equally distant from the three sides.

ANSWER: FALSE

2 The circum-center of a triangle is equally distant from the three sides.

ANSWER: FALSE

3 The ortho-center of a triangle is equally distant from the three vertices.

ANSWER: FALSE

CONTEST 35 ROUND 4B: TRUE OR FALSE

1 If $\sin A = \frac{1}{2}$, then A is an acute angle

ANSWER: FALSE

2 If $\tan A = -2$, then A is an obtuse angle.

ANSWER: FALSE

3 If $\cos A = -\frac{\sqrt{3}}{2}$, then A is an obtuse angle.

ANSWER: FALSE

CONTEST 35 ROUND 5: RIDDLE

1 I am a vector.

2 As a vector, I am rather strange.

3 I have no particular direction.

4 I have a length which is nothing to write home about.

5 I am the identity for vector addition.

6 I am a constant under scalar multiplication.

Who am I?

ANSWER: ZERO VECTOR

CONTEST 36 ROUND 1A

A and B are acute angles with $\sin A = \frac{1}{2}$, and $\cos B = \frac{1}{2}$. Find

1 $\cos(A - B)$

ANSWER: $\sqrt{3}/2$

$[A = 30^\circ, B = 60^\circ, \cos(A - B) = \cos(30^\circ - 60^\circ) = \cos(-30^\circ) = \sqrt{3}/2]$

2 $\sin(A + B)$

ANSWER: 1

$[\sin(A + B) = \sin(30^\circ + 60^\circ) = \sin 90^\circ = 1]$

3 $\tan(A - B)$

ANSWER: $-1/\sqrt{3}$ or $-\sqrt{3}/3$

$[\tan(A - B) = \tan(30^\circ - 60^\circ) = \tan(-30^\circ) = -\tan 30^\circ = -1/\sqrt{3} = -\sqrt{3}/3]$

CONTEST 36 ROUND 1B

A binary operation is defined on the set Z of integers by $a * b = a^2 - ab + b^2$

Solve for x given

1 $x * 5 = 31$

ANSWER: $x = 6, -1$

$[x^2 - 5x + 25 = 31, x^2 - 5x - 6 = 0, (x - 6)(x + 1) = 0, x = 6, -1]$

2 $x * x = 3x$

ANSWER: $x = 0, 3$

$[x^2 - x^2 + x^2 = 3x, x^2 = 3x, x^2 - 3x = 0, x(x - 3) = 0, x = 0, 3]$

3 $x * (2x) = 3$

ANSWER: $x = \pm 1$

$[x^2 - 2x^2 + 4x^2 = 3x^2 = 3, x^2 = 1, x = \pm 1]$

CONTEST 36 ROUND 1C

Find constants A and B such that

1 $(x + 7)/(x - 3)(x + 2) = A/(x - 3) + B/(x + 2)$

ANSWER: A = 2, B = -1

$[x + 7 = A(x + 2) + B(x - 3), x = 3, 10 = 5A, A = 2; x = -2, 5 = -5B, B = -1]$

2 $(2x + 4)/(x + 1)(x - 1) = A/(x + 1) + B/(x - 1)$

ANSWER: A = -1, B = 3

$[2x + 4 = A(x - 1) + B(x + 1), x = 1, 6 = 2B, B = 3; x = -1, 2 = -2A, A = -1]$

3 $(x - 4)/(x + 2)(x - 1) = A/(x + 2) + B/(x - 1)$

ANSWER: A = 2, B = -1

$[x - 4 = A(x - 1) + B(x + 2), x = 1, -3 = 3B, B = -1; x = -2, -6 = -3A, A = 2]$

CONTEST 36 ROUND 2: SPEED RACE

1 Find the zeros of the function $f(x) = x^3 - 4x$

ANSWER: $x = 0, \pm 2$

$[x(x^2 - 4) = 0, x = 0, \pm 2]$

2 Express as a single logarithm $\log(x^3 + 27) - \log(x^2 - 3x + 9)$ and simplify.

ANSWER: $\log(x + 3)$

$[\log((x + 3)(x^2 - 3x + 9)/(x^2 - 3x + 9)) = \log(x + 3)]$

3 Find the period T of the trigonometric function $y = \tan(2x - \pi)$

ANSWER: $T = \pi/2$

$[\tan(2x - \pi) = \tan 2x, 2T = \pi, \text{ hence } T = \pi/2]$

CONTEST 36 ROUND 5: RIDDLE

1 I am more closely associated with a circle.

2 I am a part of a circle.

3 Mind you, I am not a sector.

4 Neither am I a segment of the circle.

5 I am as a matter of fact, a part of the circle.

6 I am a portion of the circumference of the circle.
Who am I?

ANSWER: ARC

CONTEST 36 ROUND 4A: TRUE OR FALSE

1 For the function $y = \sin^{-1}x$, the domain is $\{x: -1 \leq x \leq 1\}$

ANSWER: TRUE

2 For the function $y = \cos^{-1}x$, the domain is $\{x: -1 \leq x \leq 1\}$

ANSWER: TRUE

3 For the function $y = \tan^{-1}x$, the domain is $\{x: x \geq 0\}$

ANSWER: FALSE

CONTEST 36 ROUND 4B: TRUE OR FALSE

1 No two consecutive integers can both be prime numbers.

ANSWER: FALSE (2 and 3 are both consecutive and prime)

2 For any integer n , $n(n + 1)$ is an even number.

ANSWER: TRUE

3 For any integer n , $n(n + 1)(n + 2)$ is divisible by 3.

ANSWER: TRUE

CONTEST 37 ROUND 1A

Find the value of n if

1 $x^3z = 1$, $x^2 = y$ and $z = y^n$

ANSWER: $n = -3/2$

[$z = 1/x^3 = 1/(x^2)^{3/2} = (x^2)^{-3/2} = y^{-3/2} = y^n$, hence $n = -3/2$]

2 $x^4z^3 = 1$, $y = x^{1/2}$, $z = y^n$

ANSWER: $n = -8/3$

[$z = x^{-4/3} = (x^{1/2})^{-8/3} = y^{-8/3} = y^n$, hence $n = -8/3$]

3 $x^5/z^2 = 1$, $y = x^{-1/2}$, $z = y^n$

ANSWER: $n = -5$

[$z = x^{5/2} = 1/(x^{-1/2})^5 = 1/y^5 = y^{-5} = y^n$, hence $n = -5$]

37 ROUND 1B

1 If $\tan(A + B) = 3/2$ and $\tan B = 1/2$, find $\tan A$.

ANSWER: 4/7

$$[\tan A = \tan(A + B - B) = (\tan(A + B) - \tan B) / (1 + \tan(A + B)\tan B) = (3/2 - 1/2) / (1 + (3/2)(1/2)) = 1/(7/4) = 4/7]$$

2 If $\tan(A - B) = 1/3$ and $\tan B = 1/2$, find $\tan A$.

ANSWER: 1

$$[\tan A = \tan(A - B + B) = (\tan(A - B) + \tan B) / (1 - \tan(A - B)\tan B) = (1/3 + 1/2) / (1 - (1/3)(1/2)) = (5/6) / (5/6) = 1]$$

3 Given $\tan(B - A) = 1/2$ and $\tan B = 1/4$, find $\tan A$

ANSWER: -2/9

$$[\tan A = \tan(B - (B - A)) = (\tan B - \tan(B - A)) / (1 + \tan B \tan(B - A)) = (1/4 - 1/2) / (1 + (1/4)(1/2)) = (-1/4) / (9/8) = -2/9]$$

CONTEST 37 ROUND 1C

Differentiate the given function with respect to x .

1 $y = (x - 1/x)^5$

ANSWER: $dy/dx = 5(1 + 1/x^2)(x - 1/x)^4$

2 $y = (x^2 - 1/x^2)^8$

ANSWER: $dy/dx = 16(x + 1/x^3)(x^2 - 1/x^2)^7$

$$[dy/dx = 8(2x + 2/x^3)(x^2 - 1/x^2)^7 = 16(x + 1/x^3)(x^2 - 1/x^2)^7]$$

3 $y = (x^3 + 1/x^3)^{-5}$

ANSWER: $dy/dx = -15(x^2 - 1/x^4)(x^3 + 1/x^3)^{-6}$

$$[dy/dx = -5(3x^2 - 3/x^4)(x^3 + 1/x^3)^{-6} = -15(x^2 - 1/x^4)(x^3 + 1/x^3)^{-6}]$$

CONTEST 37 ROUND 2: SPEED RACE

1 Solve the trigonometric equation $2\sin x - 1 = 0$ for $0 < x < 2\pi$

ANSWER: $x = \pi/6, 5\pi/6$

[$\sin x = 1/2$, $x = \pi/6, 5\pi/6$]

2 If $x^2 + 2xy - y^2 = 10$, find dy/dx

ANSWER: $dy/dx = (x + y)/(y - x)$ [$2x + 2(y + xdy/dx) - 2ydy/dx = 0$,

$2(x + y) = 2(dy/dx)(y - x)$, $dy/dx = (x + y)/(y - x)$]

3 Given the functions $f(x) = \sqrt{x - 1}$ and $g(x) = x^2$, find the domain of $f(g(x))$

ANSWER: $\{x: x \geq 1, \text{ or } x \leq -1\}$

[$f(g(x)) = \sqrt{g(x) - 1} = \sqrt{x^2 - 1}$, possible for $x^2 - 1 \geq 0$, $x \geq 1$ or $x \leq -1$]

CONTEST 37 ROUND 4A: TRUE OR FALSE

1 $3x^4 - 7x^2 + 3/x$ is a polynomial of degree 4.

ANSWER: FALSE

2 $(3x^2 + 2)^4$ is a polynomial of degree 4.

ANSWER: FALSE

3 The sum of two polynomials of degree 3 is a polynomial of degree 3 or less.

ANSWER: TRUE

CONTEST 37 ROUND 4B: TRUE OR FALSE

1 The sum of the lengths of two sides of a triangle exceeds the length of the third side.

ANSWER: TRUE

2 The sum of the measures of any two angles of a triangle is greater than the supplement of the third angle.

ANSWER: FALSE

3 The sum of the interior angles of a triangle is greater than the sum of the exterior angles of the triangle.

ANSWER: FALSE

37 ROUND 5: RIDDLE

1 I am a number system.

2 I am used in computing but I am not the binary number system.

3 My digits involve the decimal digits and the first 6 letters of the alphabets.

4 With me, each digit is equivalent to a byte of 4 bits.

5 My base is an even square.

6 I have 16 digits in all.

Who am I?

ANSWER: HEXADECIMAL NUMBER SYSTEM (Accept hexadecimal)

CONTEST 38 ROUND 1A

Find the constant term in the expansion of the given binomial. You may leave answer in the form $nC_r a^r b^s$.

1 $(3x^2 + 2/x)^6$

ANSWER: $6C_2 3^2 2^4$ $[6C_r (3x^2)^r (2/x)^{6-r} = 6C_r 3^r 2^{6-r} x^{3r-6}$, for constant term $3r - 6 = 0$, $r = 2$, hence constant term is $6C_2 3^2 2^4$]

2 $(2x + 1/3x^2)^9$

ANSWER: $9C_6 (2^6/3^3)$ or $9C_3 2^6 3^{-3}$ $[9C_r (2x)^r (1/3x^2)^{9-r} = 9C_r 2^r 3^{r-9} x^{3r-18}$, $3r - 18 = 0$, $r = 6$, hence for constant term $9C_6 (2^6/3^3)$]

3 $(5x^2 + 1/2x^2)^8$

ANSWER: $8C_4 (5^4/2^4)$ or $8C_4 5^4 2^{-4}$ $[8C_r (5x^2)^r (1/2x^2)^{8-r} = 8C_r 5^r 2^{r-8} x^{4r-16}$, $4r - 16 = 0$, $r = 4$, hence constant term = $8C_4 (5^4/2^4)$]

CONTEST 38 ROUND 1B

Find the values of the scalars x and y if the resultant of the given 3 vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ is zero.

1 $\mathbf{a} = \mathbf{i} + 2\mathbf{j}$, $\mathbf{b} = (x - 3)\mathbf{i} + (x - y)\mathbf{j}$, $\mathbf{c} = (x - y)\mathbf{i} + (2 - x)\mathbf{j}$

ANSWER: $x = 0, y = -2$

$[(1 + x - 3 + x - y = 0, 2x - y = 2, 2y + x - y + 2 - x = 0, y + 2 = 0, y = -2$
 $2x = y + 2 = -2 + 2 = 0, x = 0]$

2 $\mathbf{a} = (3 - x)\mathbf{i} + (y + 1)\mathbf{j}$, $\mathbf{b} = (y + x)\mathbf{i} + x\mathbf{j}$, $\mathbf{c} = (x - y)\mathbf{i} + (5 - x)\mathbf{j}$,

ANSWER: $x = -3, y = -6$

$[3 - x + y + x + x - y = 0, 3 + x = 0, x = -3; y + 1 + x + 5 - x = 0,$
 $y + 6 = 0, y = -6]$

3 $\mathbf{a} = 2\mathbf{i} + (y - x)\mathbf{j}$, $\mathbf{b} = (x + y)\mathbf{i} + (x - 2y)\mathbf{j}$, $\mathbf{c} = -4\mathbf{i} + (2y - 4)\mathbf{j}$

ANSWER: $x = -2, y = 4$

$[2 + x + y - 4 = 0, x + y = 2; y - x + x - 2y + 2y - 4 = 0, y - 4 = 0, y = 4,$
 $x = -y + 2 = -4 + 2 = -2]$

CONTEST 38 ROUND 1C

Find the set of values of x for which

1 $2x^2 - 3x + 3 > 5$

ANSWER: $\{x: x > 2, \text{ or } x < -1/2\}$

$[2x^2 - 3x - 2 > 0, (2x + 1)(x - 2) > 0, x > 2 \text{ or } x < -1/2]$

2 $3x^2 + 4x + 1 < 5$

ANSWER: $\{x: -2 < x < 2/3\}$

$[3x^2 + 4x - 4 < 0, (3x - 2)(x + 2) < 0, -2 < x < 2/3]$

3 $4x^2 + 5x + 4 > 10$

ANSWER: $\{x: x < -2, \text{ or } x > 3/4\}$

$[4x^2 + 5x - 6 < 0, (4x - 3)(x + 2) > 0, x < -2 \text{ or } x > 3/4]$

CONTEST 38 ROUND 2: SPEED RACE

1 Given that $\sin A = 12/13$ and A is obtuse, find $\cos A$

ANSWER: $-5/13$

[angle in second quadrant hence $\cos A = -5/13$]

2 Find the measure of the angle between the vectors $\mathbf{a} = 5\mathbf{i} + 3\mathbf{j}$ and $\mathbf{b} = 3\mathbf{i} - 5\mathbf{j}$

ANSWER: 90°

[$\mathbf{a} \cdot \mathbf{b} = 15 - 15 = 0$, the vectors $\mathbf{a}$ and $\mathbf{b}$ are perpendicular, hence angle is 90°]

3 Find the value of k if the equation $x^2 - 2kx + 25 = 0$ has equal roots.

ANSWER: $k = \pm 5$

[For equal roots, $b^2 - 4ac = 4k^2 - 4(25) = 0$, $k^2 = 25$, $k = \pm 5$]

CONTEST 38 ROUND 4A: TRUE OR FALSE

1 Only rational expressions with identical denominators can be multiplied.

ANSWER: FALSE

2 $x/y \div y/x = 1$ if $x \neq 0$ and $y \neq 0$.

ANSWER: FALSE

3 A rational function is the quotient of any two real functions $f(x)$ and $g(x)$.

ANSWER: FALSE

CONTEST 38 ROUND 4B: TRUE OR FALSE

1 $\sin^{-1}(1/2) = \cos^{-1}(\sqrt{3}/2)$

ANSWER: TRUE

2 $\tan^{-1}x = \cot^{-1}(1/x)$

ANSWER: TRUE

3 $\cos^{-1}(-1/\sqrt{2}) = \tan^{-1}(-1)$

ANSWER: FALSE

CONTEST 38 ROUND 5: RIDDLE

- 1 I am a type of data.
- 2 I am data arranged in categories.
- 3 The original data is replaced with groups of data.
- 4 Frequencies are assigned to the groups.
- 5 I am a frequency distribution in which intervals replace actual data.
- 6 With me, the operative word is groups.
Who am I?

ANSWER: GROUPED DATA

CONTEST 39 ROUND 1A

- 1 The polynomial $f(x) = x^3 + ax^2 + 2x + b$ has $(x - 1)^2$ as a factor. Evaluate a and b .

ANSWER: $a = -5/2$, $b = -1/2$

$[f(1) = 1 + a + 2 + b = a + b + 3 = 0, df/dx = 3x^2 + 2ax + 2,$
 $(df/dx)_{x=1} = 3 + 2a + 2 = 0, a = -5/2, -5/2 + b + 3 = 0, b = -1/2]$

- 2 The polynomial $f(x) = x^3 + ax^2 + bx - 2$ has $(x^2 - 1)$ as a factor. Evaluate a and b .

ANSWER: $a = 2$, $b = -1$

$[(x - 1), (x + 1)$ are factors, $f(1) = 1 + a + b - 2 = 0, a + b = 1, f(-1) =$
 $-1 + a - b - 2 = 0, a - b = 3, 2a = 4, a = 2, b = 2 - 3 = -1]$

- 3 The polynomial $f(x) = ax^3 + 2x^2 + bx + 4$ has $(x + 1)(x - 2)$ as a factor. Evaluate a and b .

ANSWER: $a = -4$, $b = 10$

$[(x + 1), (x - 2)$ are factors, $f(-1) = -a + 2 - b + 4 = 0, a + b = 6, f(2) = 8a + 8 + 2b + 4$
 $= 0, 8a + 2b = -12, 4a + b = -6, 3a = -12, a = -4, b = 10]$

CONTEST 39 ROUND 1B

Find the exact increase in the area of a square if the length of a side is increased from

1 5.00 cm to 5.20 cm

ANSWER: 2.04 cm²

$$[\text{Increase in area} = (5.20)^2 - (5.00)^2 = 10.20 \times 0.20 = 2.04 \text{ cm}^2]$$

2 10.00 cm to 10.30 cm

ANSWER: 6.09 cm²

$$[\text{Increase in area} = (10.30)^2 - (10.00)^2 = 20.30 \times 0.30 = 6.09 \text{ cm}^2]$$

3 15.00 cm to 15.50 cm

ANSWER: 15.25 cm²

$$[\text{Increase in area} = (15.50)^2 - (15.00)^2 = 30.50 \times 0.50 = 15.25 \text{ cm}^2]$$

CONTEST 39 ROUND 1C

Find the first four terms in ascending powers of x in the expansion of the given binomial expression.

$$1 \quad (1 + 2x)^{1/2}$$

$$\text{ANSWER: } 1 + x - \frac{1}{2}x^2 + \frac{1}{2}x^3$$

$$[1 + \frac{1}{2}(2x) + \frac{1}{2}(-1/2)/2(2x)^2 + \frac{1}{2}(-1/2)(-3/2)/6(2x)^3 = 1 + x - (\frac{1}{2})x^2 + (\frac{1}{2})x^3]$$

$$2 \quad (1 - 3x)^{1/3}$$

$$\text{ANSWER: } 1 - x - x^2 - \frac{5}{3}x^3$$

$$[1 + (1/3)(-3x) + (1/3)(-2/3)/2(-3x)^2 + (1/3)(-2/3)(-5/3)/6(-3x)^3 = 1 - x - x^2 - (5/3)x^3]$$

$$3 \quad (1 + 4x)^{1/4}$$

$$\text{ANSWER: } 1 + x - \frac{3}{2}x^2 + \frac{7}{2}x^3$$

$$[1 + (1/4)(4x) + (\frac{1}{4})(-3/4)/2(4x)^2 + \frac{1}{4}(-3/4)(-7/4)/6(4x)^3 = 1 + x - (3/2)x^2 + (7/2)x^3]$$

CONTEST 39 ROUND 2: SPEED RACE

1 Given the matrices $A = \begin{pmatrix} 2 & 0 \\ 3 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 3 & -1 \\ 2 & -2 \end{pmatrix}$, find $\det(AB)$

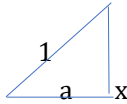
ANSWER: -8

[$\det A = 2$, $\det B = -4$, $\det(AB) = (\det A)(\det B) = 2(-4) = -8$]

2 If $\sin \theta = x$, and $0 < \theta < \frac{\pi}{2}$, find an expression for $\tan \theta$

ANSWER: $\frac{x}{\sqrt{1-x^2}}$

[$x^2 + a^2 = 1$, $a = \sqrt{1 - x^2}$, $\tan \theta = x/a = x/\sqrt{1 - x^2}$]



3 Find the equation of a curve whose gradient is given by $dy/dx = 3x^2 - 2x + 5$ and the curve passes through the point A(1, 6)

ANSWER: $y = x^3 - x^2 + 5x + 1$

[$y = x^3 - x^2 + 5x + C$, $6 = 1 - 1 + 5 + C$, $C = 1$, $y = x^3 - x^2 + 5x + 1$]

CONTEST 39 ROUND 4A: TRUE OR FALSE

The given matrix has no inverse

1 $A = \begin{pmatrix} 3 & -6 \\ 2 & -4 \end{pmatrix}$

ANSWER: TRUE

2 $B = \begin{pmatrix} 5 & -1 \\ 10 & 2 \end{pmatrix}$

ANSWER: FALSE

3 $C = \begin{pmatrix} 4 & -2 \\ 6 & -3 \end{pmatrix}$

ANSWER: TRUE

CONTEST 39 ROUND 4B: TRUE OR FALSE

1 $\log a^2 - \log a = (\log a^2)/(\log a)$

ANSWER: FALSE

2 $\log (ab)^n = (\log a + \log b)^n$

ANSWER: FALSE

3 $\log(a^3 + b^3) = 3\log a + 3\log b$

ANSWER: FALSE

CONTEST 39 ROUND 5: RIDDLE

1 In geometry, I am a point, a line, or a plane.

2 I am an object in a class.

3 I am any of the terms in the array forming a matrix.

4 I am generally a member of a collection.

5 Chemically, I am the subject of the periodic table.

6 In that context, I am sometimes classified as a metal or a non-metal.

Who am I?

ANSWER: ELEMENT

CONTEST 40 FINALS ROUND 1A

Find the value of n if

1 $\log_8 n = \frac{1}{2}p$, $\log_2(4n) = q$ and $q - p = 3$

ANSWER: $n = 8$ [$q - p = \log_2(4n) - (2/3)\log_2 n = 3$,
 $2 + \log_2 n - (2/3)\log_2 n = 3$, $(1/3)\log_2 n = 1$, $\log_2 n = 3$, $n = 2^3 = 8$]

2 $2\log_9 n = p$, $\log_3(9n) = q$ and $p + q = 6$

ANSWER: $n = 9$ [$p + q = 2\log_9 n + \log_3 9 + \log_3 n = 6$,
 $\log_3 n + 2 + \log_3 n = 6$, $2\log_3 n = 4$, $\log_3 n = 2$, $n = 3^2 = 9$]

3 $\log_5 n = (1/3)p$, $\log_5(25n^2) = q$ and $p - q = 1$

ANSWER: $n = 125$
[$p - q = 3\log_5 n - 2 - 2\log_5 n = 1$, $\log_5 n = 3$, $n = 5^3 = 125$]

CONTEST 40 ROUND 1B

Assume x and y are positive real numbers. Express y in terms of x given

1 $\sin^{-1}x = \tan^{-1}y$

ANSWER: $y = x/\sqrt{1-x^2}$ [Let $\tan^{-1}y = a$, $\tan a = y$, $\sin^{-1}x = a$,
hence $\sin a = x$, $\tan a = x/\sqrt{1-x^2}$, $y = x/\sqrt{1-x^2}$]

2 $\tan^{-1}x = \sin^{-1}y$

ANSWER: $y = x/\sqrt{x^2+1}$
[Let $\sin^{-1}y = a$, $\sin a = y$, $\tan^{-1}x = a$, $\tan a = x$, $\sin a = x/\sqrt{1+x^2} = y$]

3 $\sin^{-1}x = \cos^{-1}y$

ANSWER: $y = \sqrt{1-x^2}$
[Let $\cos^{-1}y = a$, $\cos a = y$, $\sin^{-1}x = a$, $\sin a = x$, $\cos a = \sqrt{1-\sin^2 a} = \sqrt{1-x^2}$
Hence $y = \sqrt{1-x^2}$]

CONTEST 40 ROUND 1C

Evaluate the limit

$$1 \quad \lim_{x \rightarrow 9} \frac{(x-9)}{(\sqrt{x}-3)} \quad \text{ANSWER: 6}$$

$$\left[\lim_{x \rightarrow 9} \frac{(x-9)}{(\sqrt{x}-3)} = \lim_{x \rightarrow 9} \frac{(x-9)(\sqrt{x}+3)}{(x-9)} = \lim_{x \rightarrow 9} (\sqrt{x}+3) = 3+3=6 \right]$$

$$2 \quad \lim_{x \rightarrow 25} \frac{(5-\sqrt{x})}{(x-25)} \quad \text{ANSWER: -1/10}$$

$$\left[\lim_{x \rightarrow 25} \frac{(5-\sqrt{x})}{(x-25)} = \lim_{x \rightarrow 25} \frac{(5-\sqrt{x})}{(\sqrt{x}-5)(\sqrt{x}+5)} = \lim_{x \rightarrow 25} \frac{-1}{(\sqrt{x}+5)} = \frac{-1}{(5+5)} = \frac{-1}{10} \right]$$

$$3 \quad \lim_{x \rightarrow -2} \frac{(x^3+8)}{(x+2)} \quad \text{ANSWER: 12}$$

$$\left[\lim_{x \rightarrow -2} \frac{(x^3+8)}{(x+2)} = \lim_{x \rightarrow -2} \frac{(x^2-2x+4)}{1} = (4+4+4) = 12 \right]$$

CONTEST 40 ROUND 2: SPEED RACE

1 A fair coin and a fair die are tossed once. Find the probability of obtaining a head and a number greater than 3, or a tail and a number less than 4.

ANSWER: $\frac{1}{2}$

$$[A = \{(H, 4), (H, 5), (H, 6), (T, 1), (T, 2), (T, 3)\}, P(A) = 6/12 = \frac{1}{2}]$$

2 Find the equation for the function g(x) which is obtained after the graph of the function y = 2x is reflected in the line y = x and translated 2 units to the right.

ANSWER: g(x) = $\frac{1}{2}(x-2)$ or g(x) = $\frac{1}{2}x - 1$

$$[\text{reflect in } y = x \text{ gives } x = 2y \text{ or } y = (\frac{1}{2})x, 2 \text{ units right gives } y = (\frac{1}{2})(x-2)]$$

3 If the point A(1, 6) is a point of inflexion of the curve y = x³ + ax² + bx + 2, find the values of the constants a and b.

ANSWER: a = -3, b = 6 [dy/dx = 3x² + 2ax + b, d²y/dx² = 6x + 2a,

$$\text{for } x = 1, 6 + 2a = 0, a = -3, \text{ and } 6 = 1 - 3 + b + 2, b = 6]$$

CONTEST 40 ROUND 4A: TRUE OR FALSE

For the logarithm of a positive number,

1 the mantissa is positive,

ANSWER: TRUE

2 the characteristic is positive

ANSWER: FALSE

3 the mantissa is itself a logarithm.

ANSWER: TRUE

CONTEST 40 ROUND 4B: TRUE OR FALSE

Let functions f and g have domains A and B respectively.

1...The function $(f + g)(x) = f(x) + g(x)$ has domain $A \cup B$

ANSWER: FALSE

2 The composite function $f(g(x))$ has domain $A \cup B$.

ANSWER: FALSE

3 The quotient function $f(x)/g(x)$ has domain $A \cap B$.

ANSWER: FALSE

CONTEST 40 ROUND 5: RIDDLE

1 I may be a number or a polynomial.

2 As a polynomial an example of me is $f(x) = (x + 2)(x - 3)$

3 As a number, an example of me is 6.

4 I am characterized by having proper factors.

5 I am also a purposely designed engineering material created by combining single materials to produce a structural property not present in the single materials used.

6 An example in this context is fiberglass.
Who am I?

ANSWER: COMPOSITE

